To the Graduate Council:

I am submitting herewith a thesis written by David Scott Greene entitled "Comparison Between Tapered Element Microbalance (TEOM) and Federal Reference Method (FRM) for $PM_{2.5}$ Measurement in East Tennessee." I have examined the final electronic copy of this thesis for form and content and recommend that it be accepted in partial fulfillment of the requirements for the degree of Master of Science, with a major in Environmental Engineering.

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COMPARISON BETWEEN TAPERED ELEMENT MICROBALANCE (TEOM) AND FEDERAL REFERENCE METHOD (FRM) FOR $PM_{2.5}$ MEASUREMENT IN EAST TENNESSEE

A THESIS
PRESENTED FOR THE
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ABSTRACT

Particulate matter (PM) of 2.5 microns or less is linked to serious negative health and visibility effects, and consequently is gaining increased regulatory attention. Therefore, accurate and timely measurement of ambient $PM_{2.5}$ is an essential task for both researchers pursuing productive investigations, and local air quality administrators in evaluating, modeling, planning for compliance, and generally making the highest productive use of their control resources. However, accurate and timely may be conflicting needs.

Many studies have concluded that a significant difference in measurements exists between the Tapered Element Oscillating Microbalance Monitors (TEOM) and the Federal Reference Method (FRM). The difference is generally related to lower ambient temperatures (i.e., at lower temperatures, the TEOM has been cited to under report $PM_{2.5}$ concentrations).

To determine if this same relationship exists in East Tennessee, $PM_{2.5}$ concentration and ambient temperature data were obtained from TEOM, FRM and IMPROVE sites in both an urban and a rural East Tennessee area. The urban site is a heavily developed location near downtown Knoxville, Tennessee, and the rural site is at Look Rock, located on the edge of the Great Smoky Mountains National Park overlooking the Tennessee Valley. Using TEOM and FRM data from the Knox County site and TEOM and IMPROVE data from Look Rock, comparisons were made between the 24-hr TEOM and the 24-hr

FRM/IMPROVE data for $PM_{2.5}$ concentration, and investigations were conducted to determine the effect of the 24-hr average ambient temperature on the ratio of TEOM/FRM and TEOM/IMPROVE.

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1.0 PARTICULATE MATTER

1.1 OVERVIEW

Particulate Matter (PM) is a collective term for particles, solid or liquid, found in the atmosphere. PM can be very small individual particles or an aggregate of small particles. These particles can enter ambient air through either primary or secondary pathways. Primary PM such as fugitive dust or ash is directly emitted into the atmosphere, but secondary PM is formed through the chemical reactions of other pollutants in the atmosphere; for example, when combustion gases such as SO_2, NO_X, or VOCs react with sunlight, NH_3, and/or water vapor.[1] Coagulation or agglomeration of small particles can result in larger particles, and their concentration (number of particles per unit volume) can influence this mechanism. According to Meszaros (1993), "Their small size and high concentration promote their thermal coagulation, which leads to the formation of larger particles with lower concentration."[2]

Although particulate matter can be described by its different physical and chemical properties, PM is often described and labeled by its aerodynamic diameter. PM of 10 microns (or micrometers, μm) in aerodynamic diameter or less is known as PM_{10}, and particulate matter of 2.5 microns or less is known as $PM_{2.5}$. These classifications can be further subdivided into coarse and fine particulate matter - PM between PM_{10} and $PM_{2.5}$ is categorized as coarse and $PM_{2.5}$ as fine.[3] Additionally, PM of any size below about 40-50 microns is characterized as Total Suspended Particulate Matter (TSP).[4]

PM size is a basic determinant to behavior. According to Seinfeld (1998),

> The fine and coarse particle modes, in general, originate separately, are transformed separately, are removed from the atmosphere by different mechanisms, require different techniques, for their removal from sources, have different chemical composition, have different optical properties, and differ significantly in their deposition patterns in the respiratory tract. Therefore the distinction between fine and coarse particles is a fundamental one in any discussion of the physics, chemistry, measurement, or health effect of aerosols.[5]

$PM_{2.5}$ is often found to be secondary in nature. Jacob (1999) states that particles below 1 micron in diameter, "originate almost exclusively from condensation of precursor gases."[6] Further, the Environmental Protection Agency (EPA) notes that sulfates dominate the $PM_{2.5}$ fraction in the Eastern United States.[7] Since sulfate $PM_{2.5}$ is most likely formed secondarily, it can be concluded that $PM_{2.5}$ in the east is mostly of secondary origin.

Smaller particles have higher surface to volume or surface to mass ratios; therefore, a given volume of air can exhibit a high number concentration with a large particle surface area even though it has a low mass concentration.[8]

In addition to size, PM consists of varied chemical species, and these constituents dictate behavior and impact. As an example of constituent dictated behavior, low vapor pressures of certain gases can promote condensation onto water vapor present in the atmosphere resulting in the formation of particles (secondary formation)[9]. An additional example of constituent dictated influence is the toxicity of the particles. For example, a toxic particle ingested into the body creates a more negative influence on health than does a more benignly composed particle (health concerns are further described below).

1.2 HEALTH, VISIBILITY AND CLIMATE CHANGE CONCERNS

1.2.1 HEALTH CONCERNS

PM parameters that impact human health are size and constituent. Particulate matter in general and $PM_{2.5}$ in particular are statistically associated with serious negative effects on human health.[10]

The particles themselves may be toxic, may interfere with respiratory tract mechanisms, or may allow toxic substances to be ingested by "carrying" them.[11] Significant health problems including decreased lung function, aggravated asthma, chronic bronchitis, increases in respiratory symptoms like coughing and difficult or painful breathing, and

premature death are associated with PM.[12] Research also suggests that PM can be related to cardiovascular problems.[13] Further, carbon constituents can be particularly hazardous; a recent report by the Tennessee Valley Authority concerning the Aerosol Research and Inhalation Epidemiology Study (ARIES) states that "urban particle mixes that contain high levels of carbon particles are much more likely to result in a harmful response" [14]

Particle size affects its ability for human ingestion. Relatively larger particles can be blocked by the body's nasal passages, while relatively small particles can be easily exhaled after inhalation. Particles of aerodynamic diameter of 2.5 microns or less have a lung deposition value of 50% whereas larger particles tend to be removed in the upper respiratory system by impaction or gravitational separation through the natural breathing functions.[15] Literature indicates that as size decreases, the detrimental impact of particulate matter tends to increase.[16, 17] However, both $PM_{2.5}$ and PM_{10} can penetrate into sensitive areas of the respiratory tract.

Size is not the only consideration when addressing the negative health impacts of PM. According to the National Research Council, "…size categories have been considered to have biological significance, although characteristics other than size are likely to be relevant to determining their toxicity."[18]

Even when not directly inhaled, PM may follow other pathways to the body after it settles on ground or water due to exposure to the skin or ingestion and other more indirect means.

In addition to human health effects, particulate matter can also damage materials and vegetation.[19]

1.2.2 VISIBILITY CONCERNS

PM can decrease visibility through the light scattering properties of particles. A direct relationship exists between light scattering and fine particle mass; furthermore, fine particles are more effective at scattering light than coarse particles.[20] Malm *et al*. assert that fine PM is the major cause of visibility reduction.[21] The Environmental Protection Agency states that particulate matter in general is the "major cause of reduced visibility in many parts of the country." [22]

1.2.3 CLIMATE CHANGE CONCERNS

While much scientific effort, public attention and regulatory ambition is focused on the role of gases in climate change, aerosols have been recognized as an influence on climate. Relative to "greenhouse gases," atmospheric residence time of aerosols is lower, diurnal effect is different, and their distribution is likely more regional (or less diffuse); however, aerosols can behave in ways that serve to cool the planet's surface.[23] According to Seinfeld , "Aerosols influence climate directly by the scattering and absorption of solar radiation and indirectly through their role in cloud condensation nuclei."[24] This behavior

is related to the PM constituent, chiefly sulfates and organic carbon. Sulfate condensation can enlarge clouds or extend their lifetime, and organic carbon can absorb solar radiation due to its dark color,.[25, 26]

1.3 FEDERAL REGULATORY OVERVIEW

The developing recognition of the importance of $PM_{2.5}$ in ambient air has been followed with an associated evolving regulatory scrutiny. EPA's standards for total suspended particulate matter have existed since 1969, and the PM_{10} standard was adopted in the 1987 revision of the Clean Air Act Admendments.[27,28]

Standards for criteria pollutants further evolved when the EPA revised the National Ambient Air Quality Standards (NAAQS) in 1997. Based upon public health concerns related to $PM_{2.5}$ mass concentrations, the agency determined that the pollutant should be addressed as a separate class of pollutant than PM_{10}; therefore, a new standard was developed for $PM_{2.5}$. The standard set forth by the agency was an annual concentration of 15 $\mu g/m^3$ and a 24-hour concentration of 65 $\mu g/m^3$.[29]

The addition of $PM_{2.5}$ to the NAAQS criteria pollutant list was not without controversy. In particular, EPA estimated the costs of attainment at $8.6 billion (and an estimated benefits range of $19 billion to $104 billion).[30] Prior to adding this pollutant to the criteria list and thus subjecting it to regulatory oversight and management, EPA conducted an extensive review to determine the relationship between public welfare and $PM_{2.5}$.[31]The data review revealed strong, positive statistical correlations between $PM_{2.5}$

concentrations and public health parameters. EPA is mandated by the United States Congress to promulgate NAAQS regulation based on public welfare, and, in EPA's view, these associations were sufficient to warrant $PM_{2.5}$'s placement on NAAQS[32]

However, the addition of $PM_{2.5}$ to the NAAQS has been debated. The statistical associations relied upon were not biological cause and effect, but stochastic based on judgments regarding such aspects as the strength, consistency, specificity, and coherence of the association. The Air & Waste Management Association's (AWMA) Critical Review of the $PM_{2.5}$ standard found disagreements among researchers regarding the interpretation of the epidemiological data used to associate $PM_{2.5}$ with negative health effects. The crux of the debate was which $PM_{2.5}$ attributes were most closely associated with health problems rather than generic $PM_{2.5}$ mass concentration. These attributes were identified as: size, chemical composition, and quantity of problematic particles.[33] Indicative of the debate is the following by Watson(1997):

> …although North American epidemiological studies consistently find positive and significant associations between particles and health, these associations are not as consistent or strong in recent European epidemiological studies.[34]

Litigation challenging the EPA's ability to establish the standard followed. However, in a 2001 decision in the Whitman v. American Trucking Associations case, the United

States Supreme Court affirmed the EPA's interpretation of its public health-based standards which ultimately allowed the establishment of the standard.[35] The final $PM_{2.5}$ implementation rule is to be set in Fall 2004.[36] Designation of non-attainment areas for is scheduled for November 2004.[37]

Although the health concerns were judged sufficiently compelling to lead the EPA to issue new ambient air quality standards, it was recognized that greater understanding of $PM_{2.5}$'s health effects was required. In response to this uncertainty, Congress directed an independent National Research Council study to "identify the most important research priorities relevant to setting particulate matter standards, to develop a conceptual plan for particulate-matter research, and over 5 years, to monitor research progress toward improved understanding of the relationship between particulate matter and health."[38] That relationship is still elusive. In the Council's third report which evaluated current research, the council concluded, regarding assessment of hazardous particulate matter components, "There does not yet appear to be a systematic, sustained plan for implementing studies of human chronic exposure, including examining ultrafine particles."[39] In the Fourth volume released in 2004, the Council reports that while, "…many studies to date have focused primarily on total particulate mass, rather than more detailed particle characteristics, such as their chemical composition."[40]

The Council is critical of the underlying assumption necessary for addition of $PM_{2.5}$ to NAAQS which is that the health effects of these particles are independent of other pollutants.[41]

As an indication of both the lack of definitive knowledge as well as the importance to human health, EPA recently awarded its largest ever scientific research grant to study the connection between PM and health. The grant, awarded to the University of Washington is valued at $30 million dollars for a ten year period.[42]

1.4 AMBIENT MONITORING

As a NAAQS criteria pollutant, monitoring standards were established and promulgated in 40 CFR 53 and 40 CFR 58.

Given the relative lack of understanding regarding the complex formation and behavior of ambient $PM_{2.5}$, proper design and specified objectives of measurement systems is vital. The National Research Council regards ambient monitoring as critical for both air quality management and health research.[43]

1.5 SAMPLING TECHNIQUES

A variety of methods exist for measuring PM concentrations from ambient air. Useful sampling classifications are mass-only versus speciation, and continuous versus non-continuous.

Mass-only sampling involves measurement of the total mass of a particulate matter per unit volume based upon sampling particles less than or equal to a specified aerodynamic diameter. The determination of mass concentration relies on direct or indirect methods.

The goal of this method is to determine the mass concentration of the particulate matter regardless of its chemical composition. By contrast, chemical speciation involves determining the chemical constituents of the particulate matter. An assortment of techniques is available for speciation of the chemical constituents of particulate matter. Application of these techniques is dictated by the constituents to be analyzed. As stated previously, controversy exists regarding treatment of $PM_{2.5}$ as a public health concern, and, as addressed in a previous section, both mass and its chemical constituents exercise influence over health.

Mass-only PM samplers can also be classified as Batch or Continuous. Batch instruments collect ambient data over a specified period of time, after which it is analyzed. Relative to continuous instruments, this period can be long and, therefore, not in "real time." By contrast, continuous sampling involves monitoring mass concentrations at much closer intervals so that, the measurements are in effect "real time" relative to batch measurements.

Continuous measurements are more desirable for a variety of reasons. Advantages include model validation, source apportionment, health studies and a better indication and understanding of dispersion and episodes. For example, as stated by Chung et al., "With this detailed time resolution [minutes], it would be possible for regulatory agencies to monitor peak concentration events that could then be correlated to health effects data."[44] For the local air quality administrator, accurate and timely $PM_{2.5}$ measurements can mean more effective use of control resources. Further, given that regulatory agencies have a

public health mandate regarding $PM_{2.5}$, these advantages are straightforwardly aligned with their public welfare mission. For the researcher, these advantages can mean more productive research. In addition to behavior and impact, continuous monitoring can lead to a better portrait of particulate matter levels. Arnold *et al*. estimated that using a every-sixth-day schedule could easily have overlooked the ten highest daily concentrations of PM_{10}.[45]

Examples of commercially available and commonly employed continuous $PM_{2.5}$ monitors are described in a later section.

1.6 SPECIATION METHODS

A variety of methods exist for determination of $PM_{2.5}$ chemical constituents, and the method is dictated by the constituents to be analyzed. Some common analytical methods include X-Ray Fluorescence (XRF), Proton Induced X-Ray Emission (PIXE), Inductively Coupled Plasma Emission Spectroscopy (ICP), Atomic Absorption Spectroscopy, and Gas Chromatography. None of these methods is typically real-time in the sense of providing knowledge of PM components immediate to sampling, although recent advances have produced several instruments now capable of sampling sulfates and nitrates. These are still very expensive and not as common as continuous mass samplers. Speciation methods involve obtaining either an exposed filter or a captured ambient air sample and transport to a laboratory for examination. The speciation analysis is subsequent to the sampling and hence delayed. The more recent developments

incorporate an analytical technique into the sampler, but these typically still analyze batch samples in an effort to accumulate sufficient mass to maintain accuracy.

In addition to the speciation analysis method, the monitoring device employed prior to analysis can influence the result.

1.7 MASS-ONLY DEVICES

Common mass-only monitoring devices, one non-continuous--the Federal Reference Method and two continuous--the Beta Attenuation Monitor, and the Tapered Element Oscillating Microbalance (TEOM) are described below.

1.7.1 FEDERAL REFERENCE METHOD

When EPA established $PM_{2.5}$ as a criteria pollutant, the agency established a measurement method for regulatory purposes. The Federal Reference Method (FRM) is codified in 40 CFR 50 as the regulatory standard to demonstrate attainment with regard to $PM_{2.5}$ NAAQS.

Operationally, this method collects $PM_{2.5}$, sampled at a rate of 1 m^3/hr, on a filter which is then removed, dried and weighed to obtain $PM_{2.5}$ mass concentration for a particular day.[46] The method provides regulatory satisfactory results; however, it suffers from a time lag of several days to as long as 3-4 months in cases where chemical analyses are also conducted and from possible inaccuracies. The time lag can pose difficulties in

forecasting and monitoring casual effects in brief $PM_{2.5}$ episodes and times of unstable weather patterns. Possible inaccuracies can arise from volatization of particulate matter particularly in the drying or transportation processes.[47, 48, 49, 50] In addition, post-sampling/pre-weighing adsorption of semi-volatile gases can occur.[51] It has been noted that EPA terms the approach as a "method" and not a "standard" due to these complexities in mass of particles composed of a variety of vapor pressures and combinations.[52] However, from a management perspective since the Federal Reference Method is used to determine if the standard is violated, it can, by default, be viewed as the standard method.

Another "gravimetric" sampler is the dichotomous sampler. The dichotomous sampler provides measurement of both fine and coarse PM. Utilizing the standard initial sample rate of 1 m^3/hr (16.7 L/min) through a PM_{10} size selective inlet, the flow is subsequently split into two flows: one at 90% and the other at 10% of the initial rate. Fine particles follow the higher rate flow path and the larger particles the other. These particles are then collected on separate filters which can be later weighed to determine mass concentration.[53] Although this device qualifies as a federal equivalent reference for PM_{10}, it does not qualify as a reference for $PM_{2.5}$ due to differing flow rate, filter size, and impactor type.[54]

1.7.2 BETA ATTENUATION MONITOR

The Beta Attenuation Monitor indirectly calculates $PM_{2.5}$ mass by measuring the absorption, by PM, of beta radiation on a filter. This calculation is based on the

assumption that radiation attenuation and mass are related in a consistent manner. Air is pumped through a size selective inlet to a tape filter where the PM is deposited. Based upon a time sequence, the filter is placed between the monitor's beta radiation source (^{14}C or ^{85}Kr) and a detector. The attenuation difference between the clean filter and the exposed filter reveal the mass, and combined with the constant flow rate, mass concentration can be calculated. [55] Continuous measurement can be achieved, however, the accuracy of the instrument is significantly improved by measuring the mass accumulated over a 24 hour period. Multi-day sampling can be accomplished by use of an automated tape advance mechanism that allows the tape to advance on a predetermined cycle or if too much particulate matter is deposited, resulting in an unacceptable filter resistance.

Research experience in comparing the BAM with FRM has been positive.[56, 57]

1.7.3 TAPERED ELEMENT OSCILLATING MICROBALANCE (TEOM)

The Tapered Element Oscillating Microbalance (TEOM) employs an oscillating tapered glass tube (the tapered element) which is clamped at one end and fitted with filter on the other end. As air is pulled through the tube at a constant rate, particles collect on the filter. The mass gained from the accumulated particles decreases the oscillation frequency in a identifiable manner based upon the known spring rate of the tube. From this change, mass can be determined and, applying the flow rate, mass concentration can be calculated.

The device is illustrated in Figure 1-1. The ambient air enters an EPA standard PM-10 size selective inlet. Because this standard dictates the flow rate of 1 m^3/hr (16.7 L/min), and the tapered element assembly dictates a lower flow rate, the air is isokinetically split into a 3 L/min sample stream directed to the filter and the remainder (a 13.7 L/min stream) is bypassed. To measure $PM_{2.5}$ concentrations, a sharp cut cyclone is placed immediately downstream of the PM_{10} size selective inlet and prior to the isokinetic flow splitter. Upon exit from the splitter, air flows through the inlet tube heater, to drive away moisture and to maintain a consistent temperature, and then through a 2.5 cm diameter glass fiber filter. The mass deposited on the sampler decreases the oscillating tapered element's frequency, which allows a mass concentration to be calculated. [58] Because the tapered element is very sensitive to mass changes, continuous monitoring on a real time basis is possible.

Upstream of the filter, the incoming air is heated via an inlet tube that is maintained at a constant temperature, typically 50°C, to avoid measurement problems associated with moisture or thermal expansion of the tapered element. The temperature can also be set at 30°C[59]. Although the choice of temperature can be made by the user, the default temperature provided by the manufacturer is 50 ^{o}C .[60]

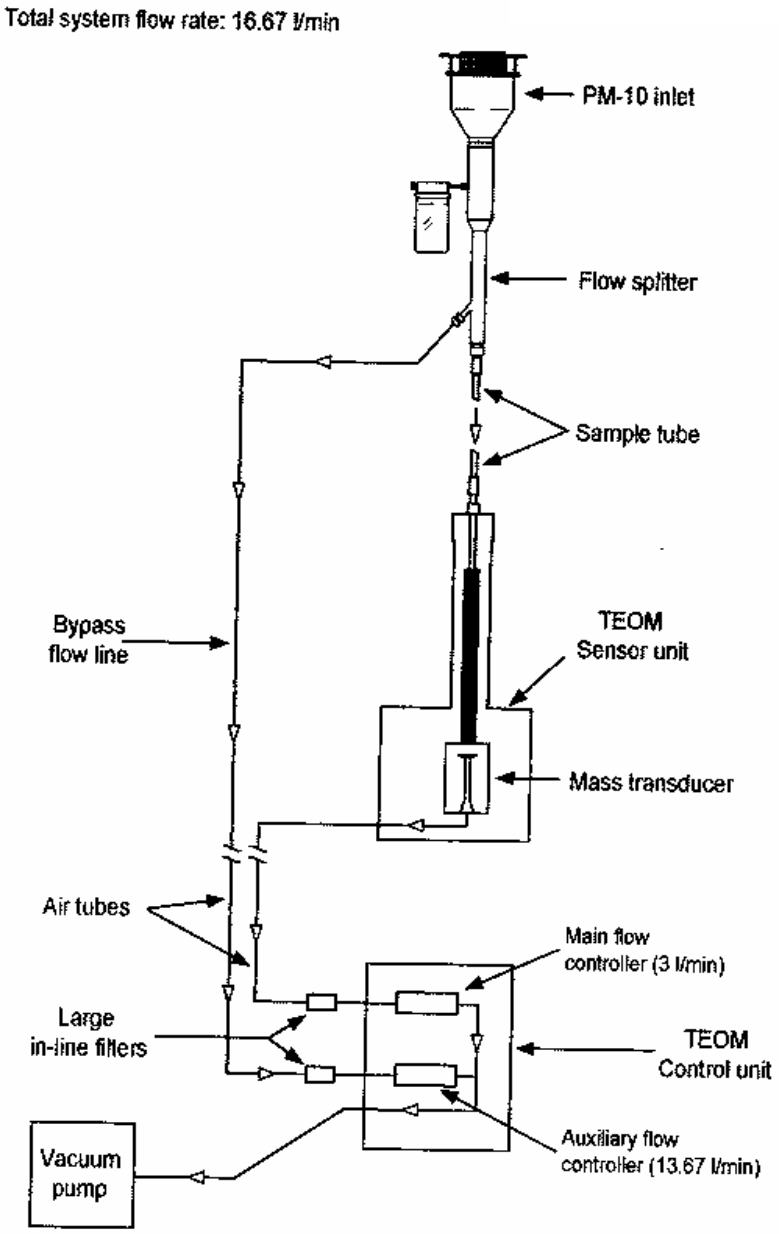


Figure 1-1: TEOM Schematic
(Source: Rupprecht & Patashnick Co., Inc.)

Establishing proper inlet tube temperature is significant because the measurement of fine particles by the TEOM can be directly affected chiefly by measuring particle-bound water and/or volatile compounds that are adsorbed on the particulate matter.[61,62, 63,64, 65] Particle bound water can be a significant mass factor when measuring fine PM.[66,67] Much debate over the use of TEOM involves its heating of ambient air. This controversy is examined in greater detail in a subsequent section.

The TEOM is manufactured exclusively by Rupprecht and Pataschink, Co., Inc. of Albany, New York, and selected specifications are listed in the Appendix.

1.8 THE TEOM/FRM DISCREPANCY AND ITS BASIC THEORY

Although the TEOM offers continuous mass concentration measurements, it is not certified as Federal Reference Method equivalent by the EPA.[68] The device's measurement is generally not consistent at all temperatures with the specified regulatory FRM for $PM_{2.5}$. The TEOM manufacturer issued a technical note advising users of the volatilization issue.[69] A consistent theme of the literature is the lack of agreement between the two measures (FRM versus TEOM) at low ambient temperatures.[70]

Numerous studies and anecdotal reports suggest that the TEOM does not provide satisfactory agreement with FRM measurements at low temperatures by under-estimating the actual $PM_{2.5}$ concentration.[71,72,73] Because moisture is driven-off, the disagreement between the methods is generally thought to result from the volatilization of organics and

ammonium nitrate caused by the heating of ambient air in the TEOM's inlet tube. These lost semi-volatiles are then unavailable for measurement, creating an incomplete representation of the ambient fine concentration, when compared to the FRM.[74,75]

In some areas with a high fraction of semi-volatile fine particle components, the TEOM may not be well suited. According to the California Air Resources Board, the TEOM has not performed well in the San Joaquin Valley Air Basin and the South Coast Air Basin; these are areas where "volatile components make up a large part of the PM."[76] This is corroborated by Interagency Monitoring of Protected Visual Environments (IMPROVE) data of ammonium nitrate portions of fine particle mass.[77]

A variety of organizations have developed correction factors and different ways to interpret TEOM readings to compensate for this problem.[78]

1.9 METHODS TO RESOLVE

Techniques to resolve the discrepancy and/or to account for the possible loss of semi-volatiles involve both post-sampling "corrections" and sampling technology applied to the TEOM. Rizzo et al. described forcing agreement by applying a correction factor to the TEOM measurements to indirectly account for lost semi-volatiles.[79] Rizzo and his co-authors are careful to point out that this factor applies only to their examined sites located primarily in the Northeast and Midwest. Graphs illustrating correlations and the effect of ambient temperatures are shown in Figure 1-2.

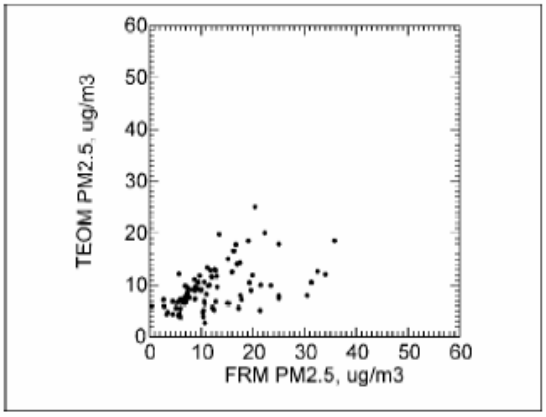


Figure 3. Uncorrected TEOM vs. FRM for site 271230868.

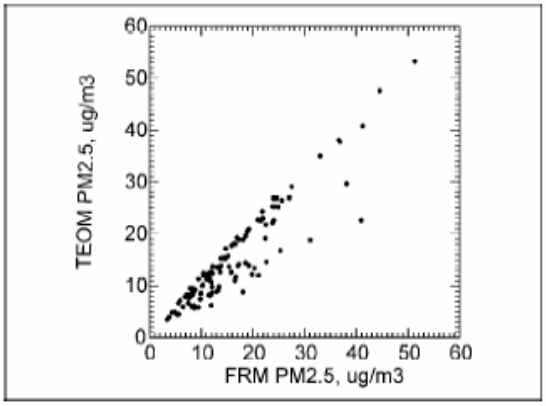


Figure 4. Uncorrected TEOM vs. FRM for site 390610040.

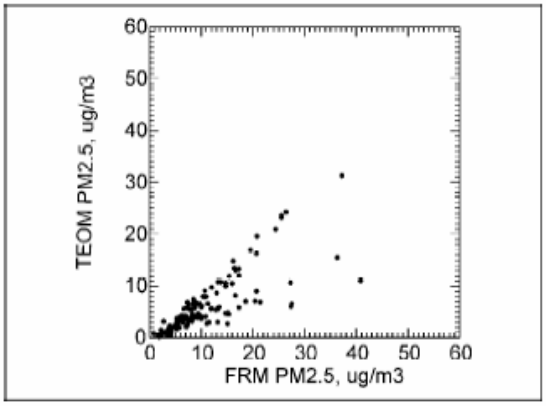


Figure 5. Uncorrected TEOM vs. FRM for site 550270007.

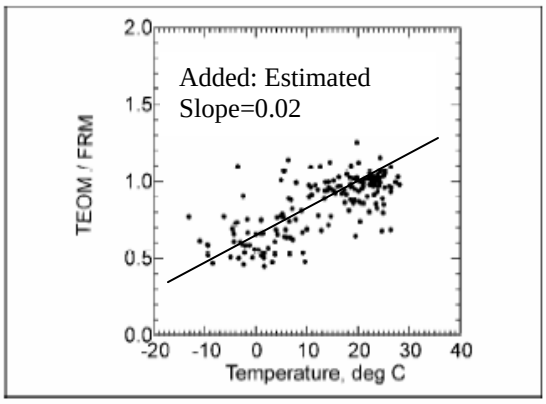


Figure 6. Ratio of TEOM to FRM for site 180970078.

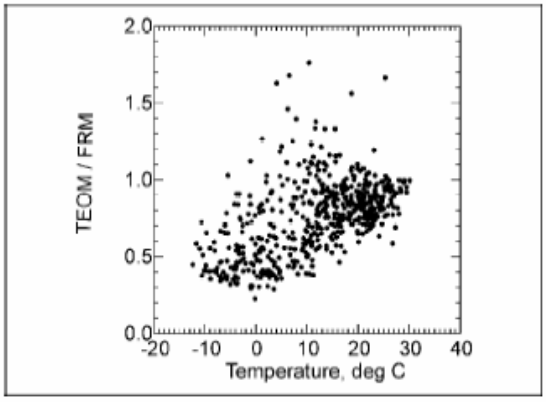


Figure 7. Ratio of TEOM to FRM for site 260810020.

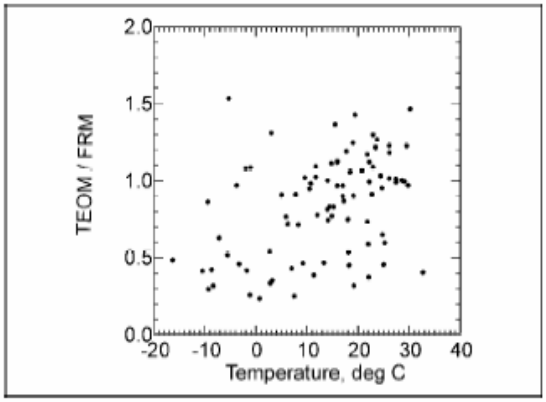


Figure 8. Ratio of TEOM to FRM for site 271230868.

Figure 1-2: Examples from Rizzo, et al., Journal of Air & Waste Management (2003) (NOTE: All TEOMs represented maintained inlet tube temperature at 50°C)

Technology applied to the TEOM is also employed. The TEOM's manufacturer, Rupprecht and Pataschink, has designed at least two approaches to dealing with the loss of semi-volatiles. These approaches are concerned more with preventing the loss of semi-voltaile mass and thus achieving a more highly accurate measure than with matching FRM measurements.

The Sampler Equilibration System (SES) is a Rupprecht & Pataschnick technique that employs the use of nafion dryers, mounted upstream of the element, combined with a lower inlet tube temperature (inlet tube temperature closer to ambient temperature). The reduced temperature, 30°C serves to maintain more of the semi-volatile compounds than at 50°C while the nafion dryer membrane removes water vapor. A heated inlet tube remains necessary to remove water in the liquid phase.[80] The Southeast Aerosol Research and Characterization (SEARCH) study employs this method in its network which is located in a relatively warm and humid climate.[81,82] However, research has shown that the system may still exhibit a negative bias.[83]

Another technique from the manufacturer is the Filter Dynamics Measurement System (FDMS) which dual samples ambient air: once through a heated inlet tube like a traditional TEOM and once through an unheated inlet. Mass subtraction of the unheated sample from the heated sample results in a estimate of the semi-volatiles which would ordinarily be lost in a traditional TEOM set-up.[84] The system essentially measures the volatile and semi-volatile components separately. The manufacturer reports correlations of approximately 0.98 for its tests.[85] However, researchers at an FDMS site in

Indianapolis, Indiana found that on “warm days in normally cold months can lead to higher FDMS values relative to FRM.”[86]

Eatough et al., have tested the Real-Time total Ambient Mass Sampler (RAMS) which is a modified dual-inlet (monitor and blank) TEOM that uses a “sandwich filter to capture semi-volatiles. The “sandwich filter” is a Teflon –coated filter followed by a charcoal impregnated filter which adsorbs all gas-phase compounds. The data are then corrected for the blank measurement, which accounts for gas-phase species captured on the noon-blank, and a particulate concentration which includes semi-volatiles can be measured.[87,88,89]

2.0 $PM_{2.5}$ IN THE EAST TENNESSEE VALLEY

2.1 INFLUENCES ON AIR QUALITY

East Tennessee is the setting of this study. Topographical features can exercise an impact on the movement of air, and therefore, airborne $PM_{2.5}$ is also influenced by these features. Likewise, the climate of the East Tennessee Valley can also exert influence on the behavior of fine particulates. Federal $PM_{2.5}$ area designations criteria allow states to include factors such as topography and meteorology when determining area boundaries.[90]

2.1.1 TOPOGRAPHY OF THE EAST TENNESSEE VALLEY

The East Tennessee Valley is one of the three major geographic divisions within the state of Tennessee. The location and major features are shown in Figure 2-1. Surrounding the Tennessee River, the East Tennessee division is bounded on the west by the Cumberland Plateau an escarpment of elevation ranging from approximately 400 meters (1300 feet) to over 800 meters (2600 feet), and bounded on the east by the Unaka/Great Smoky Mountains (Southern Appalachians) with a top elevation of 1890 meters (6200 feet) near Knoxville. Valley elevations range from approximately 700 to 200 meters (2300 to 650 feet). The eastern boundary serves as the state border between Tennessee and North Carolina. The Valley is oriented southwest to northeast. This orientation combined with the elevation differences generally directs and constrains wind patterns. Figure 2-2 reveals a wind rose for Look Rock and Figure 2-3 shows a wind rose for a lower elevation. The length of the Valley is approximately 240 kilometers (150 miles).[91]

East Tennessee Valley Region

Knoxville



Look Rock

Figure 2-1: East Tennessee Valley
(Source: ESRI, NGS)

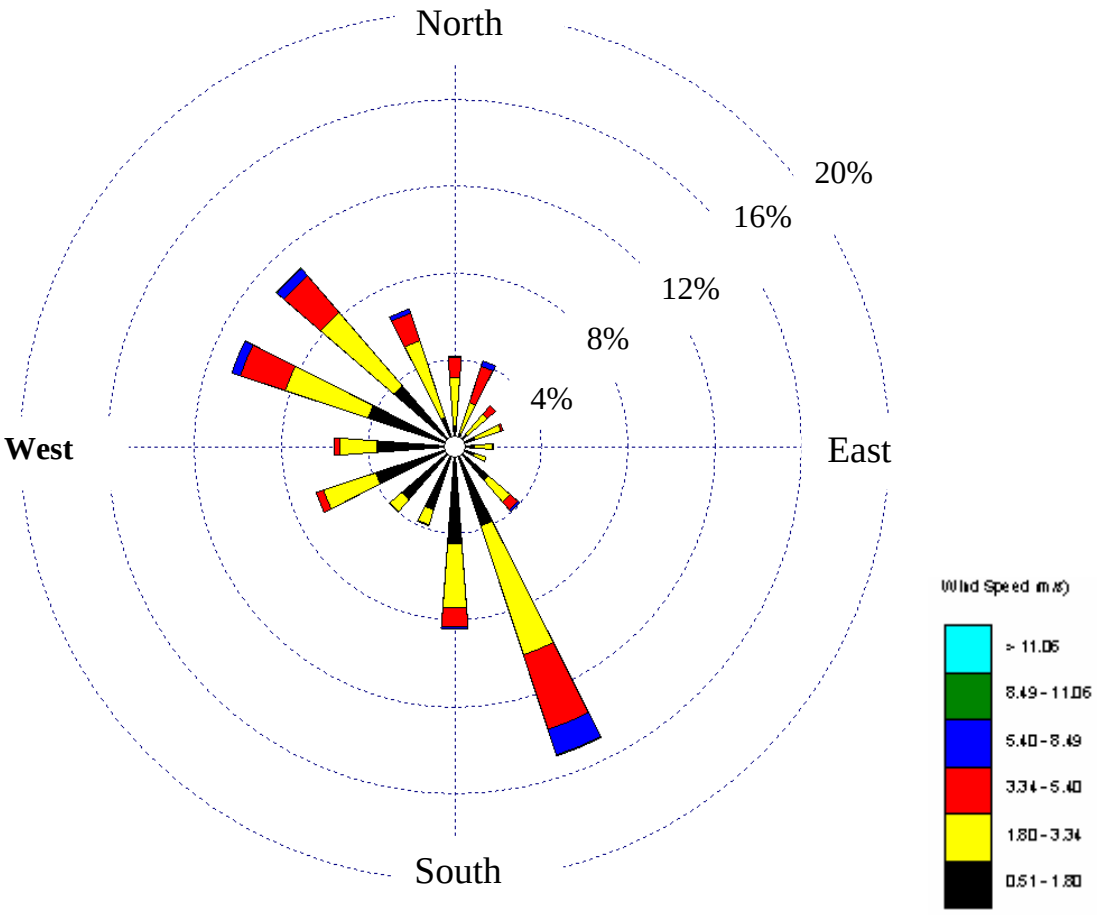


Figure 2-2: Wind Rose for Higher Elevation
(Look Rock January 2003 – December 2003)

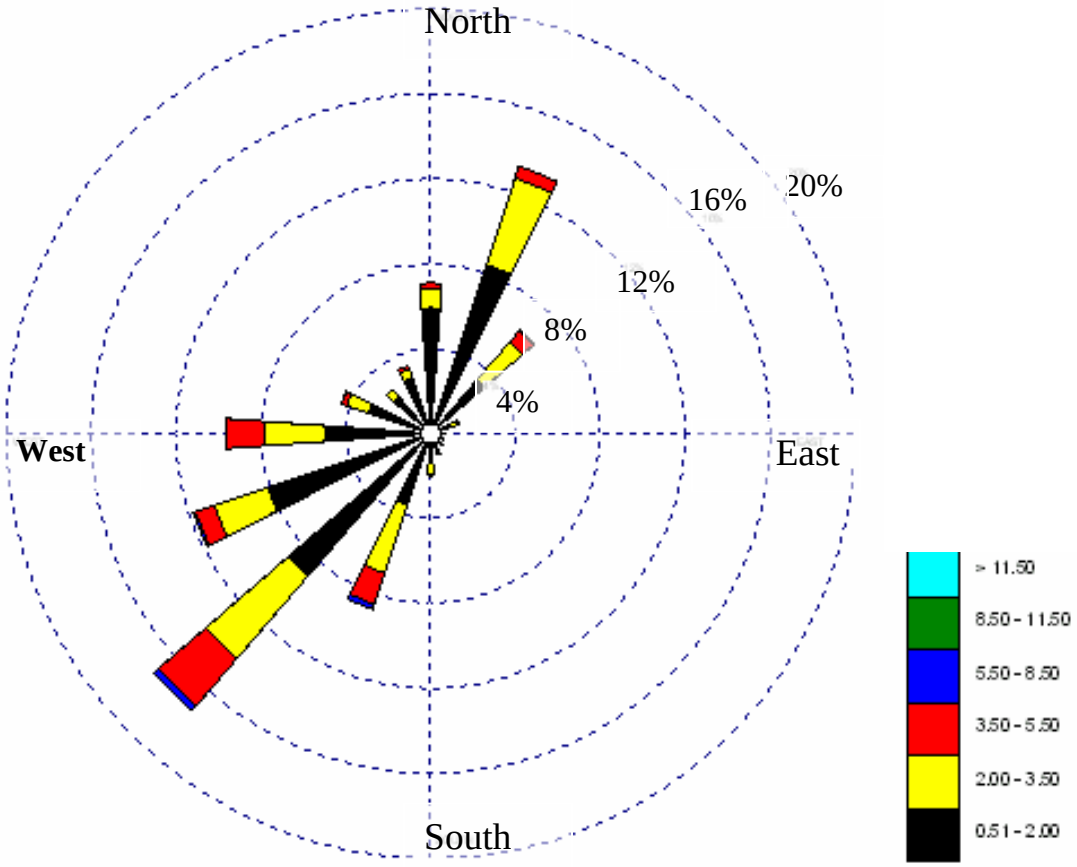


Figure 2-3: Wind Rose for Lower Elevation
(I-40/Watt Road Interchange September 2003 – August 2004)

2.1.2 CLIMATE OF THE EAST TENNESSEE VALLEY

The scale addressed is mesoscale, and the Knoxville area serves as a proxy for East Tennessee weather patterns. Normal seasonal temperature and precipitation patterns for the Knoxville area are shown in Table 2-1. Temperature and precipitation patterns for the period under study, 2002 -2003 are shown in Table 2-2.

Generally, seasonal variation is more pronounced for $PM_{2.5}$ than for PM_{10}.[92] This variation can be due to a variety of factors including meteorology, power production from combustion sources, solar radiation available and additional aspects conducive to the formation of secondary $PM_{2.5}$. Seasonal $PM_{2.5}$ variation for Look Rock and the Knox County Air Lab is shown in Figure 2-4. $PM_{2.5}$ concentrations rise during spring and summer and decrease during fall and winter.

2.2 SOURCES

A variety of anthropogenic and biogenic sources influence $PM_{2.5}$ concentrations at Look Rock and the Knox County Sites

2.2.1 PRIMARY

Primary sources for $PM_{2.5}$ include roads, agriculture, fugitive dust, possible forest fires, and industry. A large aluminum furnace and rolling facility, which is one of Tennessee's

Table 2-1: Normal Seasonal Weather Patterns

	Knoxville Area		
	Avg. Temp (^{o}F)	**Avg. Temp (^{o}C)**	**Precipitation (in)**
Jan	37.6	3.11	4.57
Feb	41.8	5.44	4.01
Mar	49.7	9.83	5.17
Apr	57.8	14.33	3.99
May	66.0	18.89	4.68
Jun	73.8	23.22	4.04
Jul	77.7	25.39	4.71
Aug	76.9	24.94	2.89
Sep	70.8	21.56	3.04
Oct	58.8	14.89	2.65
Nov	49.0	9.44	3.98
Dec	40.9	4.94	4.49

Source: National Weather Service

Table 2-2: 2003 Weather Patterns

	Knoxville Area		
	Avg. Temp (^{o}F)	**Avg. Temp (^{o}C)**	**Precipitation (in)**
Jan	41.8	5.4	3.17
Feb	40.9	4.9	8.69
Mar	53.3	11.8	2.07
Apr	60.4	15.8	5.95
May	67.5	19.7	8.09
Jun	72.1	22.3	2.45
Jul	76.3	24.6	8.27
Aug	77.8	25.4	5.48
Sep	69.3	20.7	5.14
Oct	58.5	14.7	1.24
Nov	53.0	11.7	4.71
Dec	38.4	3.6	3.41

Source: National Weather Service

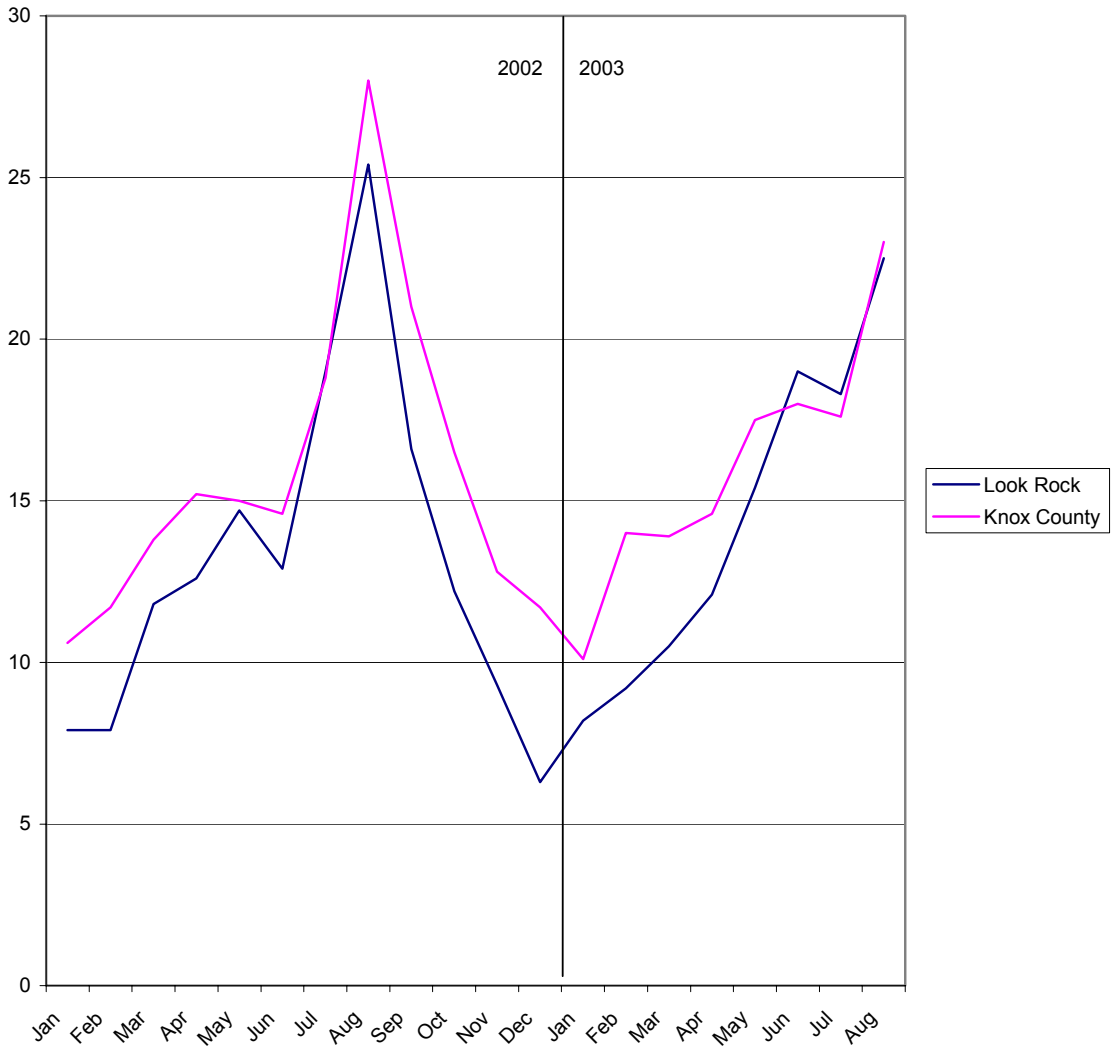


Figure 2-4: $PM_{2.5}$ Concentration Seasonality (TEOM data)

10 largest emitters of primary $PM_{2.5}$ is located approximately 16 kilometers (10 miles) from Look Rock.[93] Also located in proximity to the Look Rock site is a 870 MW coal-fired electric plant (TVA's Bull Run plant) which is also one of the state's 10 largest emitters of primary fine PM.[94,95] Other coal-fired power plants located in the region include TVA's John Sevier (712 MW) and Kingston (1456 MW) plants.[96] As a relatively high elevation site, Look Rock may also be may be more exposed to long range transport of $PM_{2.5}$ or ultrafine particulates that have agglomerated than would a lower elevation site. Knox County, a lower elevation site, is exposed to primary $PM_{2.5}$ originating and exacerbated by vehicle traffic associated with nearby arterial streets serving downtown Knoxville and the confluence of two interstates, I-40 and I-75, and it is subject to a greater influence from diurnal inversions and nighttime trapping of pollutants which may explain the tendency for higher values.

2.2.2 POSSIBLE SECONDARY

As previously stated, secondary formation of $PM_{2.5}$ is a physiochemical process that alters precursor emissions. As such, determination of sources for secondary $PM_{2.5}$ is an exercise in uncertainty. Estimates require knowledge of precursor gas emitters, wind patterns, and include residence and removal times. Further, the sources may not be in the immediate vicinity of the receptor sites. Tanner and Parkhurst state that "a major portion of the fine mass in the Tennessee Valley is regionally transported."[97] Considering such uncertainty, there is a wide variety of possible secondary $PM_{2.5}$ sources are present in the vicinity of both Look Rock and Knox County Air Lab. Regional point sources of Primary $PM_{2.5}$ and regional point sources of $PM_{2.5}$ precursors SO_2, NO_x, and VOC are

shown in Figures 2-5 through 2-8, respectively. Tables in the Appendix list emissions for the top Tennessee point source emitters. Possible sources along the Southwest/Northeast for secondary formation that are further distant include metropolitan Atlanta, Georgia, metropolitan Birmingham, Alabama which includes a primary metals industry and vehicular traffic from Washington, D.C./Northern Virginia area. Sources from the West include industry from Middle Tennessee, the population center of metropolitan Nashville, and the TVA Gallatin and New Johnsonville steam plants.

2.3 SEMI-VOLATILES

The role semi-volatiles play in comparing TEOM and FRM $PM_{2.5}$ measurements is elaborated in Section 2.6. A key semi-volatile in $PM_{2.5}$ is ammonium nitrate which is formed from ammonia and nitric acid; its atmospheric formation and chemical formula is shown below:

$$NH_3(g)+HNO_3(g) \Leftrightarrow NH_4NO_3$$

This compound is often a major component of $PM_{2.5}$ and has been identified as one of the major components lost due to heating in the TEOM inlet tube.[98, 99,100] Ammonium Nitrate is formed in areas of high ammonia and low sulfate concentration.[101] Therefore, an important consideration in evaluating the ammonium nitrate component of $PM_{2.5}$, is the sulfate concentration. Park, et al., states that "The main source of sulfate and nitrate aerosols is atmospheric oxidation of SO_2 and nitrogen oxides to H_2SO_4 and HNO_3 respectively."[102] Competition for available ammonia exists between sulfate and nitrate.[103]

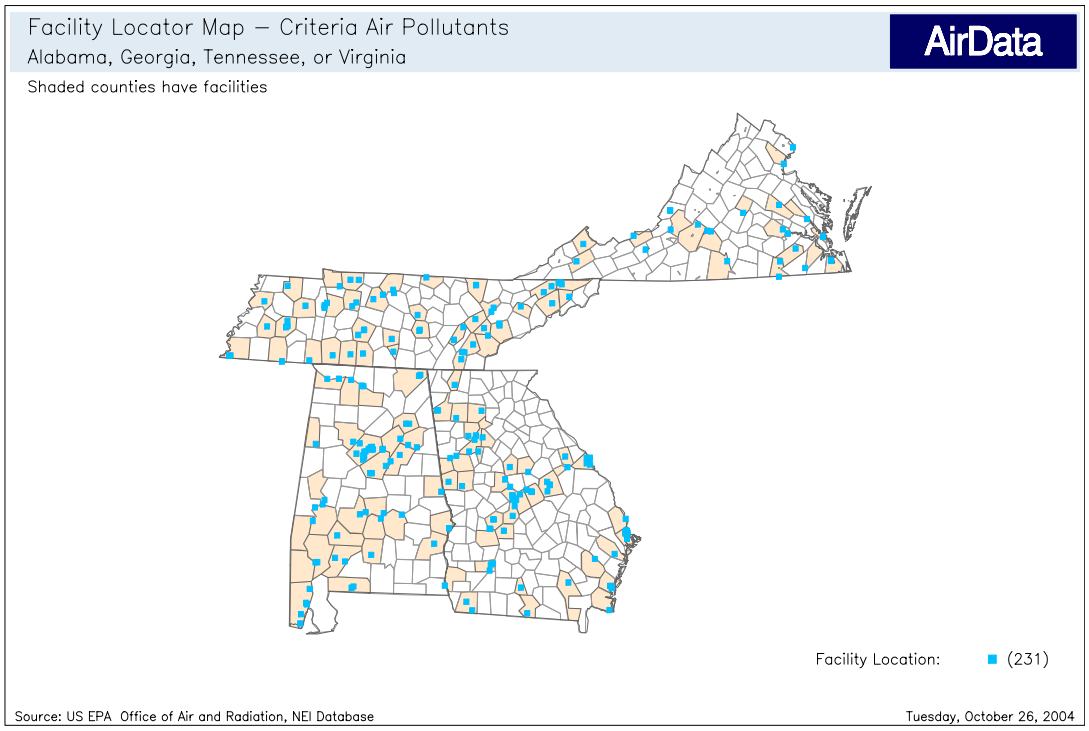


Figure 2-5: Regional $PM_{2.5}$ Point Sources

(>100Tons/yr 1999)

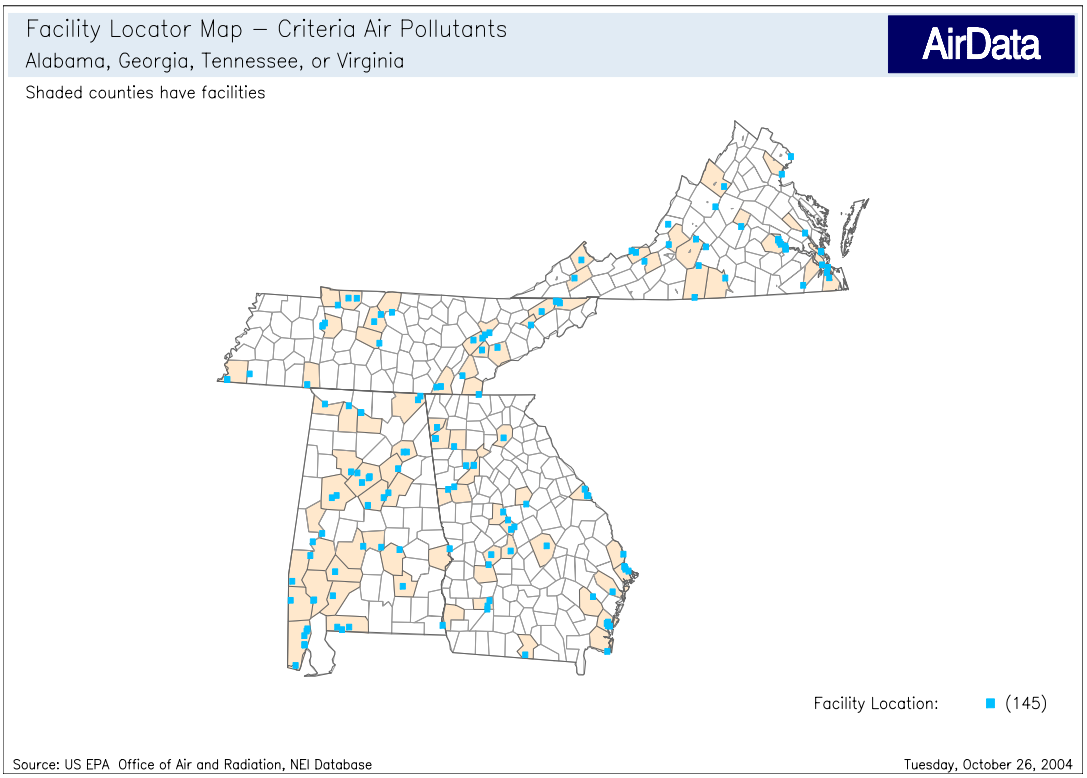


Figure 2-6: Regional SO_2 Point Sources

(>500Tons/yr 1999)

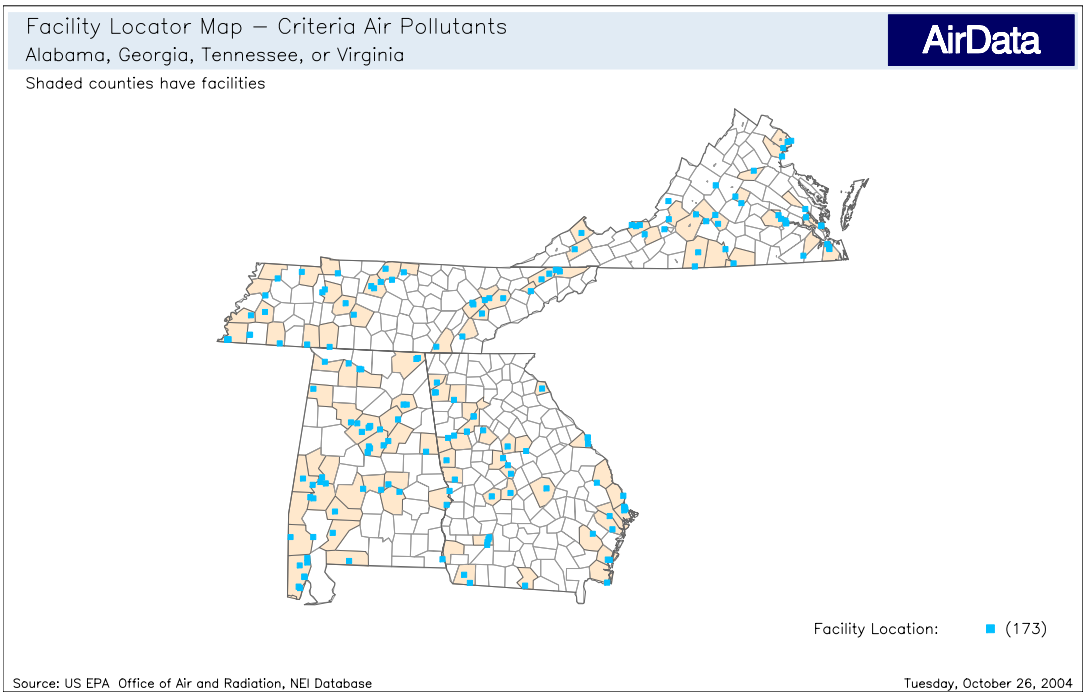


Figure 2-7: Regional NOx Point Sources

(>500Tons/yr 1999)

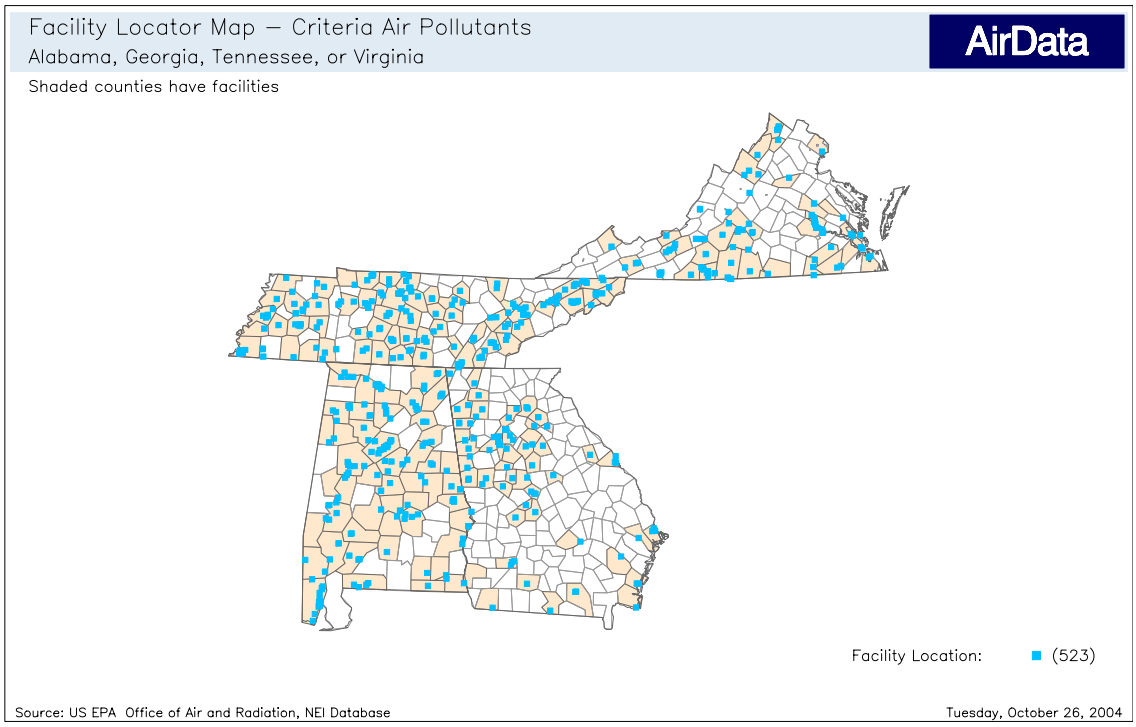


Figure 2-8: Regional VOC Point Sources

(>100Tons/yr 1999)

Although the associated dynamics are complex, the competition can be illustrated through the following reactions:

.

$$NH_3(g)+HNO_3(g) \Leftrightarrow NH_4NO_3(g)$$

$$2NH_4^+(g)+SO_4^{2-}(g) \Leftrightarrow (NH_4)_2SO_4(s)$$

$$NH_3(g)+H_2SO_4(g) \Leftrightarrow (NH_4)_2SO_4(s)$$

In the competition for the ammonium, the sulfate reaction is preferential due to sulfate neutralization.[104, 105] A table of physical properties is provided in the appendix (Table AP-4).

Since these constituents can vary regionally, and because the focus of this study is on the East Tennessee Valley, fine particulate fractions in East Tennessee are of special interest. In the Tennessee Valley, the ammonium sulfate fraction of $PM_{2.5}$ is vastly greater than the ammonium nitrate fraction. Tanner and Parkhurst estimate the chemical composition of this region's $PM_{2.5}$ to be 30-50% $(NH_4)_2SO_4$.[106] Most applicable to the Look Rock and Knox County sites, Tanner and Parkhurst's study estimated that the Knoxville area $PM_{2.5}$ composition of NH_4NO_3 was estimated to less than 10%.[107] Further, data from IMPROVE show the East Tennessee region and Look Rock in particular as having the lowest annual percentage of ammonium nitrate as a percent of Fine PM Mass in the nation.[108] These results imply that little ammonia nitrate exists as $PM_{2.5}$; This lack of ammonia nitrate is critical because the key difference between the TEOM and FRM is the loss of semi-volatiles in the TEOM's heated inlet tube. If little ammonia nitrate is present then little is available to be driven-off in the TEOM's heated inlet tube.

2.4 IMPACT OF AMMONIUM NITRATE FOR EAST TENNESSEE

If this major semi-volatile constituent of $PM_{2.5}$ is not a key constituent for a given sample, then the identified central discrepancy between the FRM and TEOM will be minimized. As can be seen in a subsequent section, the TEOM and FRM both exhibit strong correlation and no discernable effect from low ambient temperature as contrasted with Figure 1-2. This agreement could be, at least partially, explained by the relative absence of significant quantities of ammonia nitrate.

3.0 DATA

In this study, monitor data were obtained at two sites to use in comparing the TEOM observations to the FRM/IMPROVE "gravimetric" observations. Aside from having co-located monitors, the sites are dissimilar (e.g., one rural, the other urban).

3.1 LOCATIONS

3.1.1 RURAL SITE – LOOK ROCK

The rural area monitor, maintained by the National Park Service, is located at Look Rock within the Great Smoky Mountains National Park area at an elevation of 2700 feet (823 meters).[109] As with many other National Park areas, the Great Smoky Mountains is a Class I visibility area.[110] Although the location is in a protected area at a high relative elevation, lower elevation anthropogenic sources are numerous and significant as noted in Section 2.

The Rupprecht & Patashnick TEOM located at Look Rock has been in service since April 2002. The inlet tube is heated to a temperature of 30° C with upstream nafion dryers. Gravimetric $PM_{2.5}$ data is reported through the Interagency Monitoring of Protected Visual Environments (IMPROVE) network from concentrations determined by the Air Quality Group at the Crocker Nuclear Lab/University of California-Davis in Davis, California. Although not strictly defined as Federal Reference Method, 40 CFR 58 notes that the IMPROVE system, which relies upon gravimetric measurement, is acceptable as an approximation of FRM for non-compliance violations measures.[111,112]

3.1.2 URBAN SITE– KNOX COUNTY AIR LAB

Near downtown Knoxville, Knox County Air Quality Management operates a TEOM for $PM_{2.5}$ monitoring at its Knoxville Regional Laboratory. This location is in proximity to mobile sources from both arterial streets and two interstates, and point sources from nearby industry. The inlet tube of the Rupprecht & Patashnick TEOM is heated to a temperature of 50° C. An FRM data monitor is also located at the same site.

3.2 DATA COLLECTION

3.2.1 RURAL LOCATION

Look Rock IMPROVE $PM_{2.5}$ concentration data were provided by Jim Renfro, Air Resource Specialist at the Great Smoky Mountains National Park. The associated temperature data were obtained from the National Park Service Air Resources Division for the period of May 8, 2002 to May 3, 2003. TEOM continuous concentration measurements were obtained from the Environmental protection Agency's AIRnow site. TEOM concentration measurements that were invalid (e.g., monitor unresponsive or for other reasons not providing accurate measurements according to EPA's site) were not considered for subsequent calculation.

3.2.2 URBAN LOCATION

Knox County FRM $PM_{2.5}$ concentration and temperature data were provided by Chris Tunkel Laboratory Technician, Knox County Air Quality Management for the period of April 20, 2002 to April 3, 2003. TEOM continuous concentration measurements were

obtained from the Environmental Protection Agency's AIRnow site. As with the Look Rock data, TEOM concentration measurements that were invalid were not considered for subsequent calculation.

For both the rural and urban sites, TEOM concentrations were measured in micrograms/cubic meter ($\mu g/m^3$) and reported by hour. The hourly concentrations were then averaged over the 24-hour (midnight to midnight) period for a daily average concentration. For comparison, TEOM daily data were matched only to the applicable days for which FRM data existed.

3.3 RESULTS

Figures 3-1 and 3-2 are scatter plots of the TEOM versus IMPROVE/FRM concentration measurement for Look Rock and Knox County respectively.

These plots display the relationship between the measurements obtained from the different methods. The r^2 value is high for each site: 0.9606 for Look Rock and 0.9181 for Knox County. These relatively high values combined with a lack of a discernable visual pattern suggest that the monitors are highly correlated for this period.

To further investigate a temperature effect on the TEOM measurement, scatter plots were made using the ratio of the measurements (relative difference), TEOM/FRM and TEOM/IMPROVE, compared to the 24-hour average ambient temperature are shown in Figures 3-3 and 3-4.

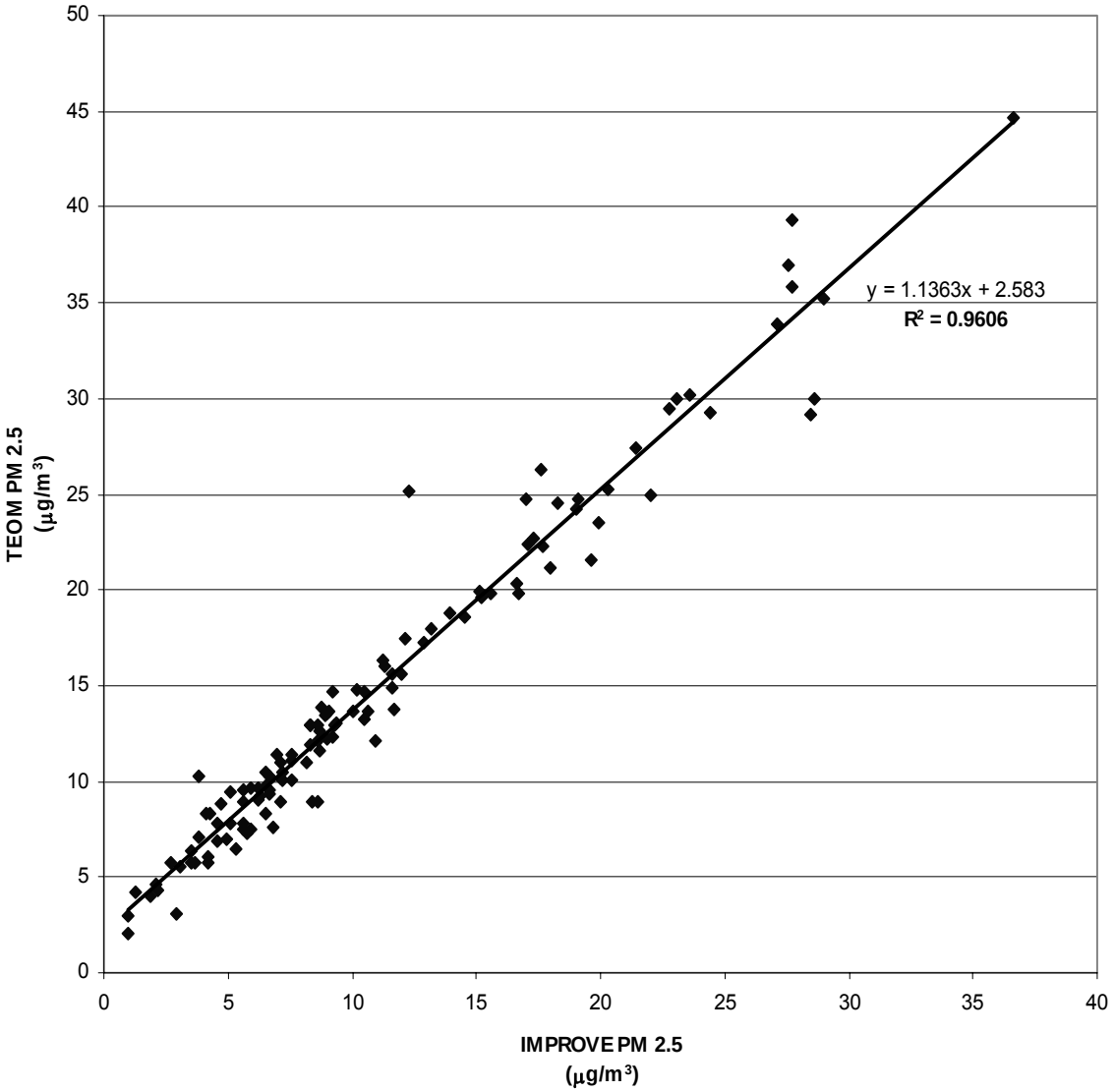


Figure 3-1: TEOM vs. IMPROVE $PM_{2.5}$ Look Rock, TN.
April 26, 2002 - April 18, 2003

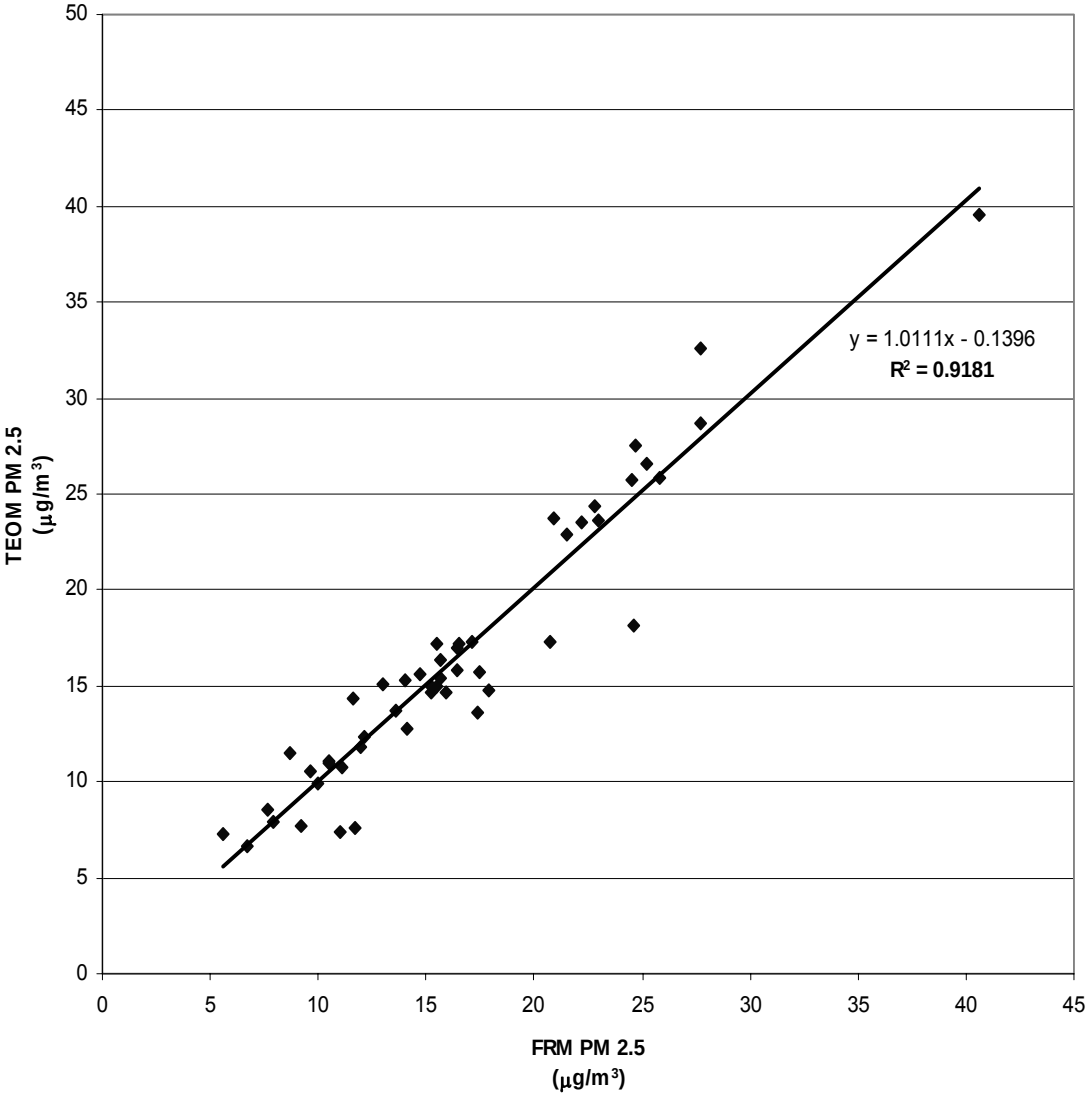


Figure 3-2: TEOM vs. FRM $PM_{2.5}$ Knox County Air Lab.
April 20, 2002 - April 3, 2003

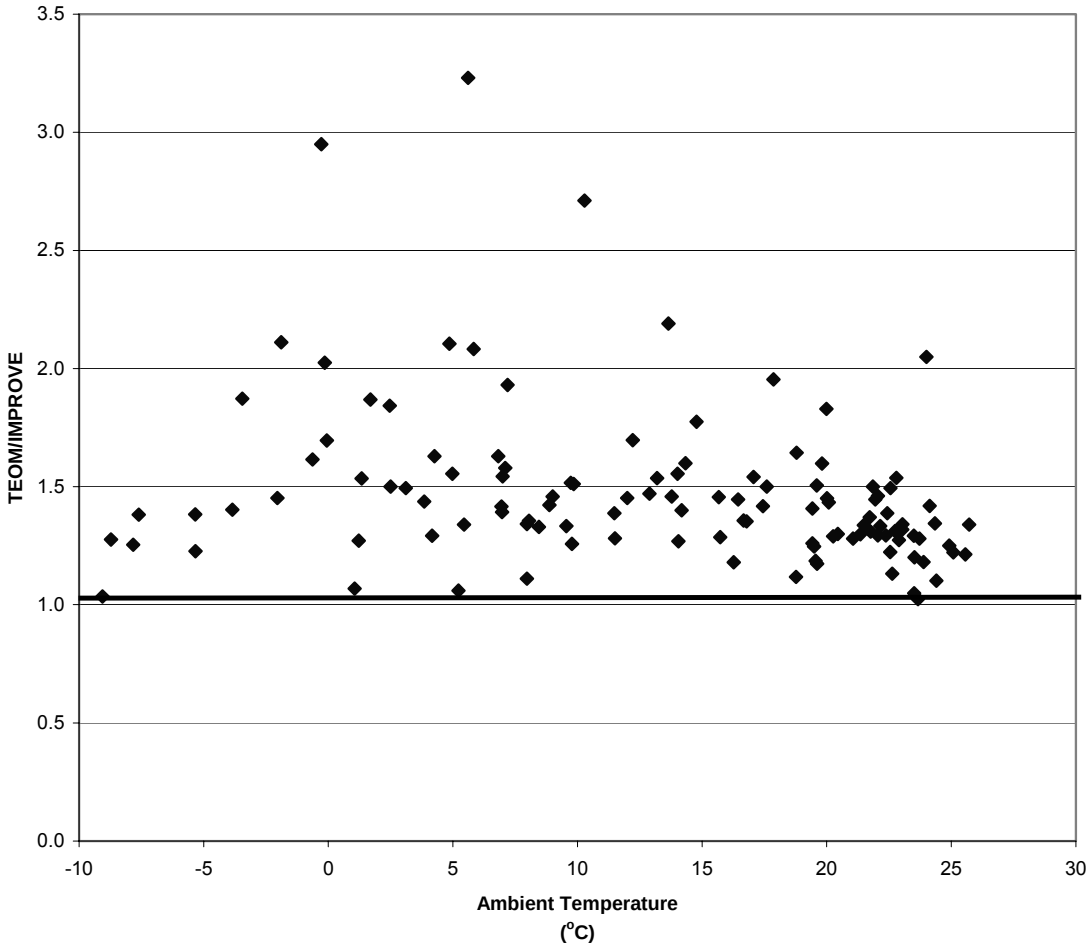


Figure 3-3: TEOM to IMPROVE Ratio Look Rock.
(April 26, 2002 - April 18, 2003)

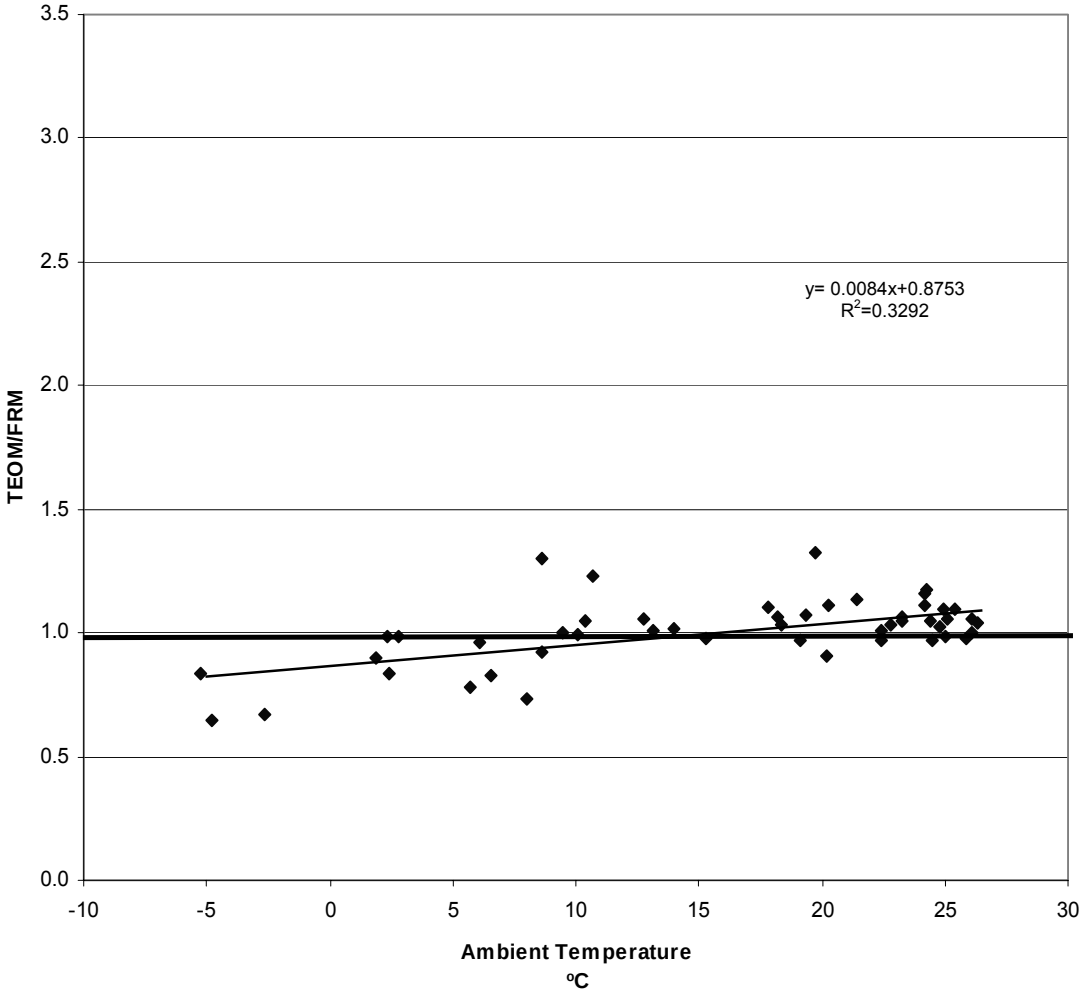


Figure 3-4: TEOM vs. FRM Ratio Knox County Air Lab.
(April 20, 2002 - April 3, 2003)

For complete agreement between the TEOM and the FRM/IMPROVE concentration measurements, the ratio results in a value of 1. However, if the TEOM measurements were systematically lower than FRM or IMPROVE values at low temperatures, low temperature points would be grouped below the ratio value of 1. Contrary to previously reported literature data, these figures do not display TEOM measurements systematically lower than those collected by the FRM or IMPROVE method. At the rural location, Figure 3-3, no pattern indicating underreporting at low temperatures is apparent. The TEOM is in close agreement, but, counter to previous studies, may actually over predict PM concentrations compared with standard gravimetric (e.g., compared to Figure 1-2).

As noted previously, the inlet tube temperature at Look Rock is maintained at 30° C as opposed to the 50° C setting of the Knox County TEOM. This temperature difference may explain or partially explain the consistently higher TEOM measurement at Look Rock. This explanation is consistent with the theory that a lower temperature setting results in higher retained concentration of volatile constituents whereas a higher temperature would possibly volatilize them.

Regarding Knox County Air Lab, Figure 3-4 shows that the beginnings of an ambient temperature effect may be apparent; two of the low temperature points are lower than any of the other data points (at a fraction of 0.650 and 0.673). However, a clear pattern, and hence an ambient temperature effect cannot be claimed to appear since the lowest temperature point is in greater agreement (0.837), and 5 of the 7 “coldest” observations exhibit an average ratio of .908. Further, discernable “improvement” with increasing

temperatures is not definitive. The urban site exhibits no significant difference between the values using Wilcoxon's method (note the very small slope of 0.0084 in Figure 3-4). This is in sharp contrast to data reported by Rizzo et al in which the slope of TEOM/FRM versus ^{o}C is approximately 0.02 (see Figure 1-2) and shows significant change in the ratio as function of ^{o}C. Although the ratio at higher temperatures are more likely to be greater than 1.0 in Figure 3-4, the slope would be insignificantly small (less than 0.01) suggesting very little if any effect of ambient temperature.

A comparison of the difference between the TEOM and gravimetric reported values (TEOM-FRM) is shown in Figures 3-5 and 3-6. A noteworthy pattern for Look Rock is revealed (Figure 3-5). First, the preponderance of data is above zero indicating a consistent positive bias for the TEOM (as does the relative difference graph in Figure 3-3). An additional pattern is that of increasing positive difference as temperature increases. The increasing difference appears to begin at around 20°C with normal dispersion about the mean between -5 ^{o}C and 20 ^{o}C. Although many variables can be contributing to this increase, a cursory examination of climate data at extremes reveals little conclusive; however, the data suggests that the bias is larger during summer months when ambient temperatures are high. As seen in the appendix Table AP-9, nine of the ten most positively biased data points occurred during the summer, between June 4, 2002 and August 24, 2002 while seven of the least biased occurred during early autumn (three occurred during the summer months). However, precipitation and relative humidity do not appear to contribute significantly to the bias in this cursory examination. Further research into this curve could yield further understanding of the increasing positive

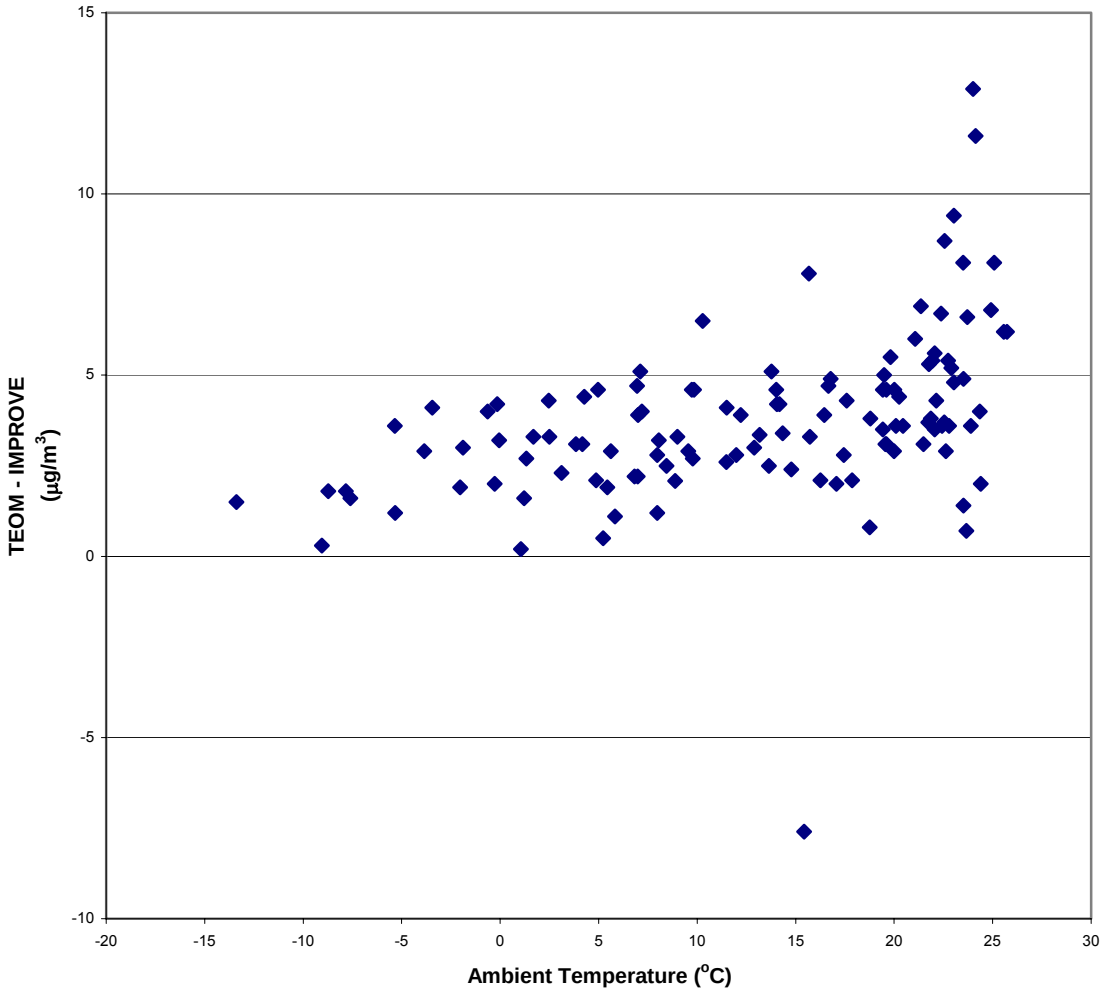


Figure 3-5: TEOM – IMPROVE Look Rock.

(April 26, 2002 - April 18, 2003)

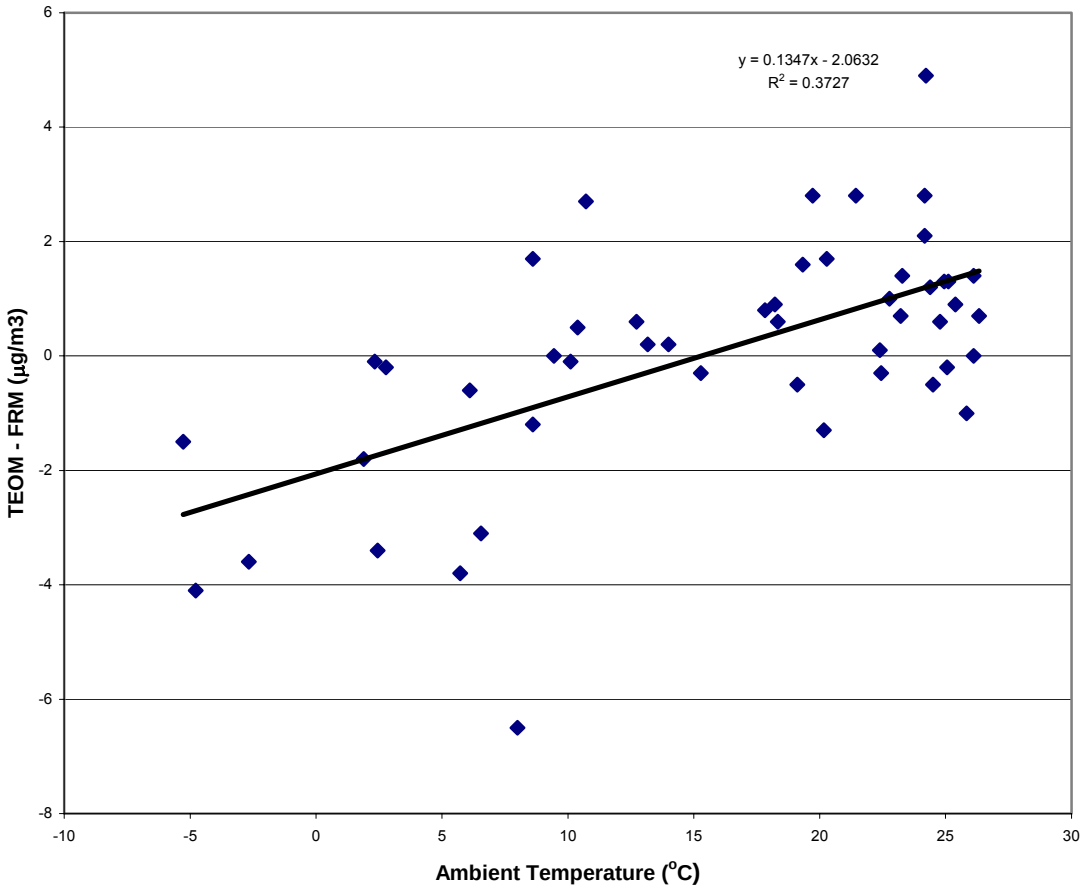


Figure 3-6: TEOM – FRM Knox County.
(April 20, 2002 - April 3, 2003)

differences shown in Figure 3-5. The corresponding graph for the Knox County site does not exhibit the same characteristics (Figure 3-6). Although the positive slope may indicate that the TEOM underreports at lower ambient temperatures the slope is low (0.13) especially when considering the lack of statistical significance in the relative difference graph of Figure 3-4. The data collected in East Tennessee show a minimal effect of temperature, particularly when compared to comparable data in Figure 1-2, where the TEOM/FRM ratio varies from 0.5 up to 1.0 as ambient temperature increases from 0°C to 30°C.

3.4 INLET TEMPERATURE DIFFERENCE

As previously stated both TEOMs examined show strong agreement with the associated FRM (or IMPROVE), and there is no significant effect due to low ambient temperature. However, at Look Rock, the "over-reporting" of TEOM relative to the IMPROVE over this period is approximately 35% percent. This data is consistent with a previous observation from the previous year when comparing Figure 3-7 to Figure 3-1.

3.4.1 MOTIVATION

As noted previously, the Look Rock inlet tube temperature is set at 30°C (with upstream nafion dryers) while the Knox County Air Lab monitor is set at 50°C. With these settings, the TEOM/ "gravimetric" both relative and absolute differences are much higher at Look Rock indicating higher $PM_{2.5}$ observations relative to the "gravimetric" monitor (IMPROVE). A hypothesis was formulated that the higher reading at Look Rock might

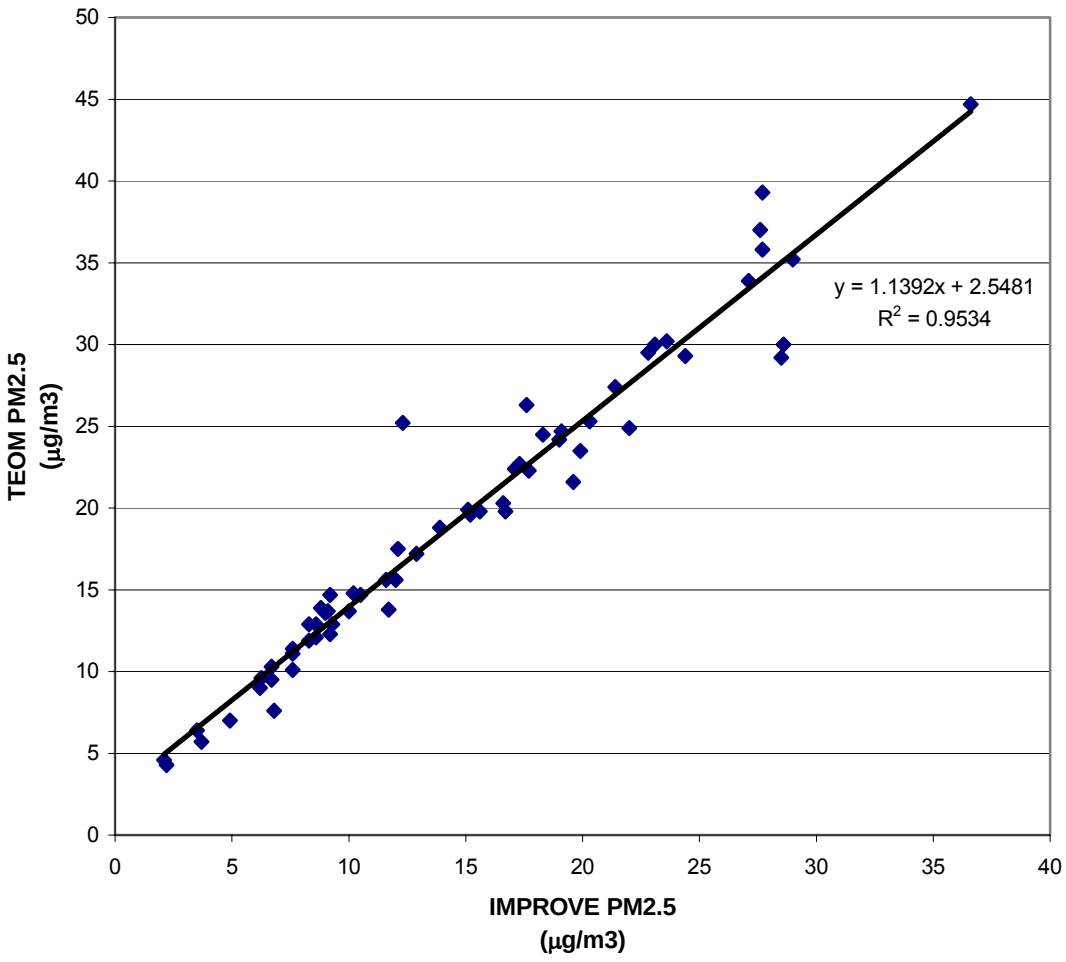


Figure 3-7: TEOM vs. IMPROVE $PM_{2.5}$ Look Rock.
(April 26, 2001 - April 29, 2002)

be the result of the lower inlet temperature (30°C), thus retaining more volatile $PM_{2.5}$ than when the 50°C inlet was used (Knox County).

To determine the impact of the temperature difference and whether $PM_{2.5}$ material is driven off by the higher temperature setting, a laboratory investigation was undertaken to determine if PM samples collected at Look Rock would de-gas if heated to 50°C. The laboratory investigation was motivated by the question of whether the TEOM inlet tube temperature differences affected the TEOM/"gravimetric" results.

3.4.2 PROCEDURAL STEPS

Ten unexposed filters were obtained from Jim Renfro, Air Quality Specialist for the Great Smoky Mountains National Park (NPS), and weighed. The box containing the unexposed filters includes a desiccant pack to reduce possible pre-use moisture content. Following this step, the filters were assigned a number, repacked and returned to the National Park Service to be exposed in the Look Rock TEOM.

From late April to July 2004, the filters were exposed to ambient $PM_{2.5}$ through use in the Look Rock TEOM. The filters were in fact used in identical manner as typical TEOM filters. Upon completion, which was dictated by the NPS determination of filter loading, the exposed filters were retrieved for further analysis.

Upon receipt, the filters were desiccated for 24 hours, and then placed in an oven at 30°C for 24 hours. After heating at this temperature to mimic the Look Rock TEOM inlet

tube, the filters were weighed and the weight recorded. To simulate the 50°C inlet tube environment, the filters were then immediately placed in a laboratory oven at 50°C. After being maintained at 50°C for 24 hours, the filters were re-weighed and their weights recorded. An equipment list is shown in the Appendix.

3.4.3 LABORATORY RESULTS

The experimental results are shown in Figure 3-8. The results do not show a consistent and commiserate reduction in the mass with increased temperature. The mass was unchanged in two observations and reduced from a low of 0.09% to a high of 0.35% with an average of 0.19%.

According to this study, there is a slight loss in mass for most (82%) of the filters, indicating that some degree of de-gassing occurred. However, the reduction was far from the 35 percent observed in the monitor data at Look Rock when the TEOM was compared to the IMPROVE data. Therefore, the inlet tube temperature difference does not appear to be the primary or one of the primary reasons for the "over-reporting" at Look Rock. At least, it was not identifiable as the cause based on post analysis.

Other possible reasons for the observation differences include an out-of-specification flow rate for the TEOM, the IMPROVE sampler (or both), or an incorrect spring rate calibration constant.

Out-of-Specification Flow-Rate: An out-off-specification flow rate could manifest itself

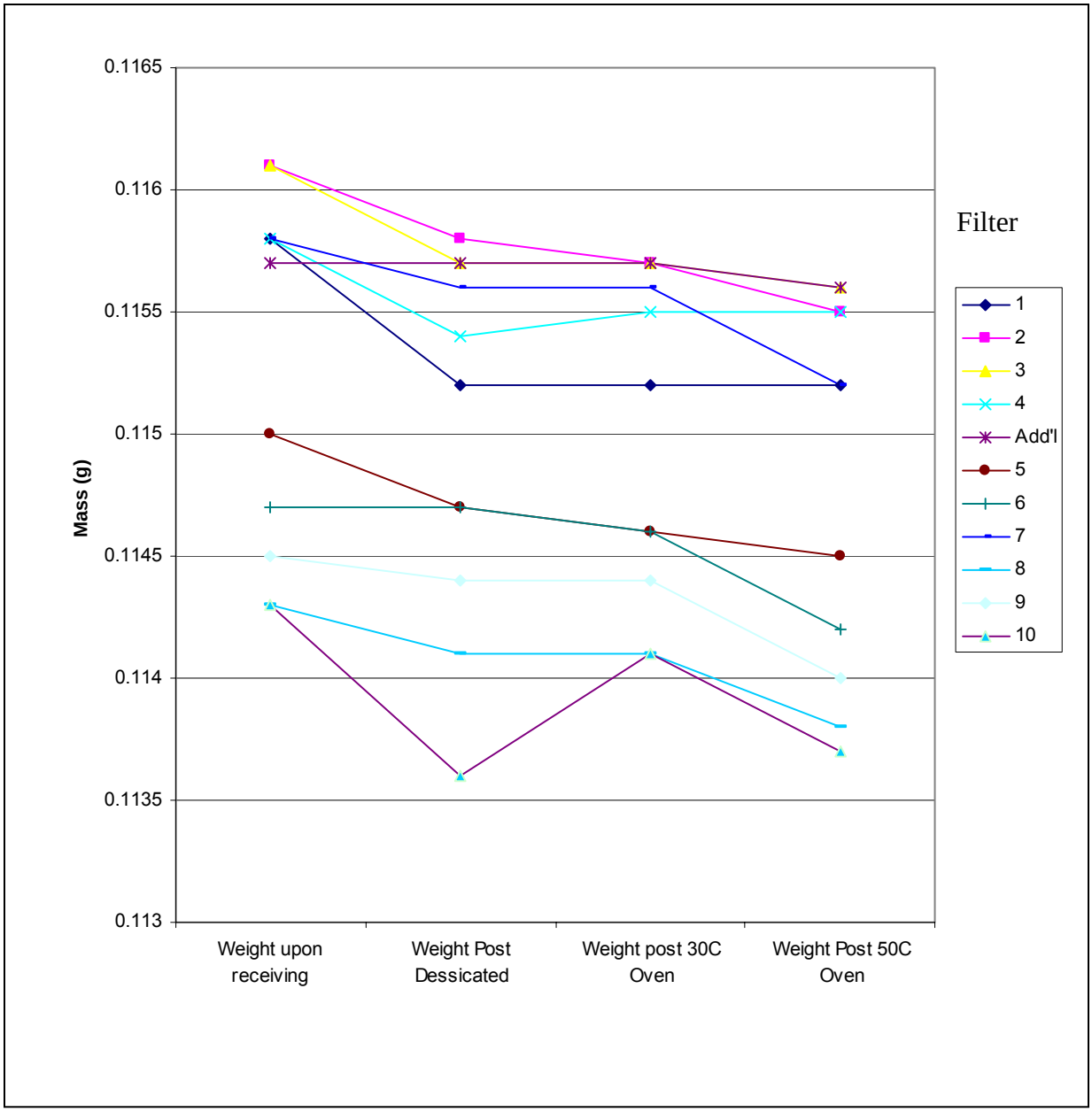


Figure 3-8: Effect of Heating Exposed Look Rock TEOM Filters

into incorrect calculation of particle mass for the TEOM. Since mass concentrations calculations are dependent on flow rate as shown below, a flow rate that is slightly lower than specification will yield a higher mass concentration and provide higher slope value.

$$\text{Mass Concentration} = \frac{\text{Smooth Total Mass}_{\text{HOUR}} - \text{Smooth Total Mass}_{\text{HOUR-1}}}{\text{Total Flow}_{\text{EPA}}}$$

(Note: Total Flow $_{\text{EPA}}$ is the volumetric flow for the period which is converted to EPA standard pressure and temperature). Likewise, the IMPROVE gravimetric sampler relies on a standard flow rate that impacts mass concentration calculation in a similar fashion (i.e., a higher flow rate will yield a lower mass concentration observation).

Incorrect Spring Calibration Constant: In the regression equation in Figure 3-1, a slope of 1.14 is observed. This suggests a possible calibration error especially when compared to the Knox County Air Lab TEOM in Figure 3-2 (1.01). The relationship between spring calibration constant and oscillation frequency relating to the change in mass is shown:

$$dm = K_o \left[\frac{1}{f_1^2} - \frac{1}{f_2^2} \right]$$

K_o is the calibration constant determined at the factory and specific to each monitor. Since measured mass is related to the calibration constant, as shown below, a high K_o would translate into inaccurately high mass observations.

These two possible explanations are not exhaustive, but suggest avenues for future research to determine the reason for over-reporting on $PM_{2.5}$ relative to IMPROVE measurements.

4.0 CONCLUSIONS AND RECOMMENDATIONS

This study examined the relationship between the Tapered Element Oscillating Microbalance (TEOM) and the Federal Reference Method (FRM)/IMPROVE in the measurement of $PM_{2.5}$ mass concentrations in East Tennessee. Conclusions drawn from this work could be used in applying continuous TEOM measurements for regulatory purposes in East Tennessee or for better understanding of how $PM_{2.5}$ mass measurement is dependent upon constituents.

The conclusions reached:

- For the monitors and time period studied, strong agreement existed between the TEOM and FRM/IMPROVE methods for measuring ambient PM2.5 mass concentrations. The r^2 value was 0.9606 for Look Rock and 0.9181 for Knox County.
- For these monitors and this period, there appears to be no significant effect of ambient low-temperature on the measurement of PM2.5 by the TEOM monitor relative to the FRM or to the IMPROVE data.
- The above conclusion stands in sharp contrast to general literature findings for areas other than East Tennessee.
- The observed agreement between the monitoring methods is possibly partly due to low amounts of a significant semi-volatile, ammonium nitrate.
- If the relative lack of ammonium nitrate accounts for most of the agreement, the TEOM method may provide $PM_{2.5}$ measurements comparable with the FRM. If measurements are comparable, then the TEOM can be easily employed in a

variety of beneficial uses for both the local air quality administrator and the researcher Examples of beneficial uses include: better understanding of high or low fine PM episodes, source apportionment, and in model validation

- The Look Rock TEOM recorded consistently higher concentrations than the IMPROVE gravimetric sampler (FRM surrogate). The TEOM measurements were approximately 35 percent higher for the period. The inlet tube temperature at Look Rock was maintained at 30° C as opposed to the 50° C setting of the Knox County TEOM. This temperature difference may explain or partially explain the consistently higher TEOM measurement. Although this explanation is consistent with the theory that a lower temperature setting results in higher retained concentration of volatile constituents whereas a higher temperature would volatilize them, results from an experimental procedure seeking to test this explanation did not show any significant de-gassing when the PM was heated to 50°C.

Reasons for such a difference invite future research. In addition to low ammonium nitrate fractions, possible reasons why these results contrast with other studies could include possible differences in climate (i.e., East Tennessee climate is not as cold as reported in the other studies), differences in geography and its effect on residence time of pollutants, differences in source types, among other possibilities. Also, research is invited to determine the reasons for the "over-reporting" of the Look Rock TEOM relative to the IMPROVE gravimetric sampler.

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APPENDIX

Table AP-1: Concentration Data for Knox County Air Lab

Day	Concentration FRM ($\mu g/m^3$)	Concentration TEOM ($\mu g/m^3$)	TEOM/FRM	Temp. (F)	Temp. (C)
04/20/02	15.7	16.4	1.045	73.8	23.2
04/26/02	10.0	9.9	0.990	50.2	10.1
05/02/02	20.9	23.7	1.134	70.6	21.4
05/08/02	21.5	22.9	1.065	73.9	23.3
05/14/02	10.5	11.1	1.057	54.9	12.7
05/20/02	10.5	11.0	1.048	50.7	10.4
05/26/02	22.8	24.4	1.070	66.8	19.3
06/01/02	22.2	23.5	1.059	77.2	25.1
06/07/02	11.1	10.8	0.973	72.4	22.4
06/13/02	24.5	25.7	1.049	75.9	24.4
06/19/02	24.7	27.5	1.113	75.5	24.2
06/25/02	14.0	15.3	1.093	76.9	24.9
07/07/02	25.2	26.6	1.056	79	26.1
07/13/02	14.1	12.8	0.908	68.3	20.2
07/19/02	12.0	11.8	0.983	77.1	25.1
07/25/02	23.0	23.6	1.026	76.6	24.8
08/06/02	25.8	25.8	1.000	79	26.1
08/12/02	40.6	39.6	0.975	78.5	25.8
08/18/02	9.6	10.5	1.094	77.7	25.4
08/24/02	16.5	17.2	1.042	79.4	26.3
08/30/02	13.0	15.1	1.162	75.5	24.2
09/05/02	15.2	14.7	0.967	76.1	24.5
09/11/02	27.7	32.6	1.177	75.6	24.2
09/17/02	27.7	28.7	1.036	73	22.8
09/29/02	15.5	17.2	1.110	68.5	20.3
10/05/02	13.6	13.7	1.007	72.3	22.4
10/11/02	15.5	15.0	0.968	66.4	19.1
10/17/02	15.9	14.7	0.925	47.5	8.6
10/23/02	17.1	17.3	1.012	55.7	13.2
10/29/02	7.7	8.5	1.104	64.1	17.8
11/04/02	7.9	7.9	1.000	49	9.4
11/10/02	8.7	11.5	1.322	67.5	19.7
12/10/02	17.9	14.8	0.827	43.8	6.6
12/22/02	5.6	7.3	1.304	47.5	8.6
12/28/02	17.5	15.7	0.897	35.4	1.9
01/03/03	6.7	6.6	0.985	36.2	2.3
01/09/03	11.6	14.3	1.233	51.3	10.7
01/15/03	11.0	7.4	0.673	27.2	-2.7
01/21/03	17.4	13.6	0.782	42.3	5.7
01/27/03	9.2	7.7	0.837	22.5	-5.3
02/02/03	16.4	15.8	0.963	43	6.1

Table AP-1: Continued

Day	Concentration FRM ($\mu g/m^3$)	Concentration TEOM ($\mu g/m^3$)	TEOM/FRM	Temp. (F)	Temp. (C)
02/08/03	11.7	7.6	0.650	23.4	-4.8
02/14/03	20.7	17.3	0.836	36.4	2.4
02/20/03	24.6	18.1	0.736	46.4	8.0
02/26/03	15.2	15.0	0.987	37	2.8
03/16/03	15.7	15.4	0.981	59.5	15.3
03/22/03	12.1	12.3	1.017	57.2	14.0
03/28/03	14.7	15.6	1.061	64.8	18.2
04/03/03	16.4	17.0	1.037	65	18.3

Table AP-2: Concentration Data for Look Rock

Day	Concentration IMPROVE ($\mu g/m^3$)	Concentration TEOM ($\mu g/m^3$)	TEOM/ IMPROVE	Temp. (F)	Temp. (C)
04/26/02	4.92	7	1.423	48.0	8.9
04/29/02	6.25	9.6	1.536	55.7	13.2
05/02/02	20.3	25.3	1.246	67.1	19.5
05/08/02	19.1	24.7	1.293	71.7	22.1
05/11/02	15.2	19.6	1.289	68.5	20.3
05/14/02	9	13.6	1.511	49.7	9.8
05/17/02	16.7	19.8	1.186	67.2	19.6
05/20/02	7.6	10.1	1.329	47.2	8.4
05/23/02	15.6	19.8	1.269	57.3	14.1
05/26/02	17.7	22.3	1.260	67.0	19.4
05/29/02	12	15.6	1.300	68.8	20.4
06/01/02	19	24.2	1.274	73.2	22.9
06/04/02	27.7	35.8	1.292	74.3	23.5
06/07/02	9.1	13.7	1.505	67.3	19.6
06/10/02	17.1	22.4	1.310	71.2	21.8
06/13/02	22.8	29.5	1.294	72.3	22.4
06/16/02	8.6	12.9	1.500	63.7	17.6
06/19/02	23.1	30	1.299	70.4	21.3
06/22/02	12.9	17.2	1.333	71.9	22.1
06/25/02	7.6	11.4	1.500	71.3	21.9
06/28/02	8.3	11.9	1.434	68.2	20.1
07/01/02	27.1	33.9	1.251	76.9	24.9
07/04/02	17.3	22.7	1.312	72.9	22.7
07/07/02	23.6	30.2	1.280	74.7	23.7
07/10/02	22	24.9	1.132	72.7	22.6
07/13/02	6.8	7.6	1.118	65.8	18.8
07/16/02	28.5	29.2	1.025	74.6	23.7
07/19/02	7.6	11.1	1.461	71.7	22.0
07/22/02	19.9	23.5	1.181	75.0	23.9
07/25/02	17.6	26.3	1.494	72.6	22.6
07/28/02	11.6	15.6	1.345	75.8	24.3
07/31/02	15.1	19.9	1.318	73.5	23.0
08/03/02	29	35.2	1.214	78.0	25.6
08/06/02	27.7	39.3	1.419	75.4	24.1
08/09/02	24.4	29.3	1.201	74.3	23.5
08/12/02	36.6	44.7	1.221	77.1	25.1
08/15/02	16.6	20.3	1.223	72.6	22.5
08/18/02	6.7	10.3	1.537	73.0	22.8
08/21/02	18.3	24.5	1.339	78.3	25.7
08/24/02	12.3	25.2	2.049	75.2	24.0

Table AP-2: Continued

Day	Concentration IMPROVE ($\mu g/m^3$)	Concentration TEOM ($\mu g/m^3$)	TEOM/ IMPROVE	Temp. (F)	Temp. (C)
08/27/02	8.6	12.1	1.407	67.0	19.4
08/30/02	9.2	12.3	1.337	70.7	21.5
09/02/02	9.3	12.9	1.387	72.4	22.4
09/05/02	19.6	21.6	1.102	75.9	24.4
09/08/02	28.6	30	1.049	74.3	23.5
09/11/02	27.6	37	1.341	73.5	23.0
09/14/02	3.5	6.4	1.829	68.0	20.0
09/17/02	21.4	27.4	1.280	69.9	21.1
09/20/02	10	13.7	1.370	71.1	21.7
09/23/02	13.9	18.8	1.353	62.2	16.8
09/26/02	2.2	4.3	1.955	64.1	17.9
09/29/02	9.2	14.7	1.598	67.7	19.8
10/02/02	12.1	17.5	1.446	71.5	21.9
10/05/02	10.2	14.8	1.451	68.0	20.0
10/08/02	11.7	13.8	1.179	61.3	16.3
10/11/02	6.7	9.5	1.418	63.4	17.4
10/14/02	6.2	9	1.452	53.6	12.0
10/17/02	8.8	13.9	1.580	44.8	7.1
10/20/02	10.5	14.7	1.400	57.5	14.2
10/23/02	8.3	12.9	1.554	57.2	14.0
10/26/02	2.1	4.6	2.190	56.6	13.6
10/29/02	3.7	5.7	1.541	62.7	17.1
11/01/02	10.6	13.7	1.292	39.5	4.2
11/04/02	3.5	5.7	1.629	44.3	6.8
11/07/02	8.3	12.9	1.554	40.9	5.0
11/10/02	5.9	9.7	1.644	65.8	18.8
11/13/02	7	11.4	1.629	39.7	4.3
11/16/02	1.3	4.2	3.231	42.1	5.6
11/19/02	8.2	11	1.341	46.3	8.0
11/22/02	4.1	8.3	2.024	31.7	-0.1
11/25/02	8.9	13.5	1.517	49.5	9.7
11/28/02	7.2	10.1	1.403	25.1	-3.8
12/01/02	9.4	13	1.383	22.4	-5.3
12/04/02	7.1	10.2	1.437	38.9	3.9
12/07/02	3.8	7.1	1.868	35.0	1.7
12/10/02	5.6	7.8	1.393	44.6	7.0
12/13/02	1.9	4	2.105	40.8	4.9
12/16/02	10.9	12.1	1.110	46.3	8.0
12/19/02	6.7	9.3	1.388	52.7	11.5
12/22/02	4.3	8.3	1.930	45.0	7.2

Table AP-2: Continued

Day	Concentration IMPROVE ($\mu g/m^3$)	Concentration TEOM ($\mu g/m^3$)	TEOM/ IMPROVE	Temp. (F)	Temp. (C)
12/25/02	4.7	8.8	1.872	25.8	-3.5
12/28/02	5.1	9.4	1.843	36.4	2.5
12/31/02	5.6	9.5	1.696	54.0	12.2
01/03/03	4.2	6.1	1.452	28.3	-2.0
01/06/03	6.5	10.5	1.615	30.9	-0.6
01/09/03	8.7	11.6	1.333	49.2	9.6
01/12/03	4.2	5.8	1.381	18.3	-7.6
01/15/03	5.3	6.5	1.226	22.4	-5.3
01/18/03	8.6	8.9	1.035	15.7	-9.1
01/21/03	8.4	8.9	1.060	41.4	5.2
01/24/03	5.8	7.3	1.259	7.9	-13.4
01/27/03	6.5	8.3	1.277	16.3	-8.7
01/30/03	2.9	3.1	1.069	33.9	1.1
02/02/03	3.8	10.3	2.711	50.5	10.3
02/05/03	2.7	5.7	2.111	28.6	-1.9
02/08/03	7.1	8.9	1.254	17.9	-7.8
02/11/03	5.9	7.5	1.271	34.2	1.2
02/14/03	5.6	7.5	1.339	41.8	5.4
02/17/03	1	3.0	2.950	31.5	-0.3
02/20/03	11.3	16	1.416	44.5	6.9
02/23/03	4.6	7.8	1.696	31.9	-0.1
02/26/03	4.6	6.9	1.495	37.6	3.1
03/01/03	9	12.2	1.356	46.5	8.1
03/04/03	7.2	10.5	1.458	48.2	9.0
03/07/03	7.1	11.0	1.543	44.6	7.0
03/10/03	6.5	9.8	1.500	36.5	2.5
03/13/03	17	24.8	1.456	60.2	15.7
03/16/03	14.5	18.6	1.282	52.7	11.5
03/19/03	3.1	5.5	1.774	58.6	14.8
03/22/03	10.5	13.2	1.258	49.6	9.8
03/25/03	13.2	17.9	1.357	62.0	16.7
03/28/03	8.7	12.6	1.446	61.6	16.4
03/31/03	5.1	7.8	1.535	34.4	1.3
04/03/03	11.6	14.9	1.286	60.3	15.7
04/06/03	6.3	9.3	1.470	55.2	12.9
04/09/03	1	2.1	2.083	42.5	5.8
04/12/03	11.2	16.3	1.458	56.8	13.8
04/15/03	18	21.1	1.174	67.3	19.6
04/18/03	5.6	9.0	1.600	57.8	14.3
Average	10.9	14.8			
Average Diff.		3.88 $\mu g/m^3$			
Difference		35.48%			

Note: Temperature data from National Park Service Air Resources Division

Table AP-3: Selected TEOM Specifications

Measurement Range	<5 mg/m3 to several g/m3
Inlet Flow Rate	16.7 L/min
Main Flow Rate	3 L/min
Filter Size	2.5 cm diameter
Sample Temperature Range	-40°C to 60°C
Inlet Temperature	50°C default
Temp. and Press. Smoothing Period	15 seconds

Table AP-4: Selected Properties

Substance Name	Molecular Formula	Molecular Weight	Decomposition (°C)	Solubility in 100 parts hot water (g/100mL)
Ammonium Nitrate	NH_4NO_3	80.04	210	241.8 (30°C)
Ammonium Sulfate	$(NH_4)_2SO_4$	132.14	-	103.1 (100 °C)
Nitric Acid	HNO_3	63.02	86	∞

Source: Perry's Chemical Engineers' Handbook (7th Edition); 1997

Note: Decomposition value for crystallize structure not atmospheric NH_4NO_3 Atmospheric NH_4NO_3 is at the gas - particle equilibrium which can disassociate at small changes in temperature or pressure (Source: Stelson, et.al., "A Note on the Equilibrium Relationship Between Ammonia and Nitric Acid and Particulate Ammonium Nitrate"; Atmospheric Environment; 13; pp 369-371; and correspondence with Russell Long, PhD; Environmental Protection Agency; December 8, 2004.)

Table AP-5: Top Primary $PM_{2.5}$ Emitters in Tennessee

Emissions (Tons)	Percent of Total Emissions	Facility Name	County
6186.679	11.21	TVA Johnsonville Fossil Plant	Humphreys
5393.338	9.77	TVA Kingston Fossil Plant	Roane
3367.437	6.1	TVA Gallatin Fossil Plant	Sumner
3244.646	5.88	TVA John Sevier Fossil Plant	Hawkins
2524.685	4.57	TVA Cumberland Fossil Plant	Stewart
2219.361	4.02	Bowater Newsprint& Directory - Calhoun Operations	McMinn
2189.549	3.97	UCAR Carbon Company Inc.	Maury
2072.143	3.75	TVA Bull Run Fossil Plant	Anderson
1609.034	2.91	Tenneco Packaging	Hardin
1586.548	2.87	Aluminum Company Of America, South Plant	Blount

Source: EPA

Table AP-6: Top SO_2 Emitters in Tennessee

Emissions (Tons)	Percent of Total Emissions	Facility Name	County
120079.1	22.22	TVA Johnsonville Fossil Plant	Humphreys
109194	20.2	TVA Kingston Fossil Plant	Roane
84909.77	15.71	TVA Gallatin Fossil Plant	Sumner
61886.66	11.45	TVA John Sevier Fossil Plant	Hawkins
38181.5	7.06	TVA Bull Run Fossil Plant	Anderson
23825.289	4.41	Eastman Chemical Company	Sullivan
15942.49	2.95	TVA Cumberland Fossil Plant	Stewart
13732.77	2.54	Allen Fossil Plant	Shelby
9260.04	1.71	Bowater Newsprint&Directory - Calhoun Operations	McMinn
7224.42	1.34	E I Dupont De Nemours & Co Inc	Davidson

Source: EPA

Table AP-7: Top NO_x Emitters in Tennessee

Emissions (Tons)	Percent of Total Emissions	Facility Name	County
82681.06	30.44	TVA Cumberland Fossil Plant	Stewart
26054	9.59	TVA Kingston Fossil Plant	Roane
21403.52	7.88	TVA Johnsonville Fossil Plant	Humphreys
17000.68	6.26	TVA Allen Fossil Plant	Shelby
13774.33	5.07	TVA -Gallatin Fossil Plant	Sumner
13531.53	4.98	TVA Bull Run Fossil Plant	Anderson
13343.44	4.91	Eastman Chemical Company	Sullivan
11778.75	4.34	TVA John Sevier Fossil Plant	Hawkins
6399.14	2.36	Bowater Newsprint&Directory - Calhoun Operations	McMinn
3611.067	1.33	Tennessee Gas Pipeline Co. (Station 860)	Hickman

Source: EPA

Table AP-8: Top VOC Emitters in Tennessee

Emissions (Tons)	Percent of Total Emissions	Facility Name	County
9767.875	10.26	Eastman Chemical Company	Sullivan
9059.47	9.51	Lenzing Fibers Corporation	Hamblen
3620.69	3.8	Ormet Aluminum Mill Products Corporation	Madison
2061.318	2.16	Bowater Newsprint&Directory - Calhoun Operations	McMinn
1589.342	1.67	World Color Press Inc.	Dyer
1588.89	1.67	Aluminum Company Of America, North Plant	Blount
1575	1.65	Norandal Usa, Inc	Carroll
1476.101	1.55	Nissan North America, Inc.	Rutherford
1419.2	1.49	Holliston Mills, Inc.	Hawkins
1295	1.36	Quebecor World	Dickson

Source: EPA

Table AP-9: Climate Data for Look Rock Highest and Lowest Absolute Difference (TEOM-IMPROVE)

(A)

Top Difference Rank	Date	Absolute Difference ($\mu g/m^3$)	Precipitation (in)	Relative Humidity (%)
1	08/24/02	12.9	0.1	68
2	08/06/02	11.6	0	64
3	09/11/02	9.4	0	58
4	07/25/02	8.7	0	83
5	08/12/02	8.1	0	64
6	06/04/02	8.1	0.1	72
7	06/19/02	6.9	0	64
8	07/01/02	6.8	0	68
9	06/13/02	6.7	0.1	73
10	07/07/02	6.6	0	63

(B)

Bottom Difference Rank	Date	Absolute Difference ($\mu g/m^3$)	Precipitation (in)	Relative Humidity
1	09/14/02	2.9	0.2	88
2	07/10/02	2.9	0	76
3	10/11/02	2.8	0.2	99
4	10/08/02	2.1	0	88
5	09/26/02	2.1	0.6	94
6	09/05/02	2	0	54
7	10/29/02	2	1.2	96
8	09/08/02	1.4	0	60
9	07/13/02	0.8	1.3	94
10	07/16/02	0.7	0	70

Source for Climate Data: National Park Service Air Resources Division

TABLE AP-10: Laboratory Equipment

Equipment Type	Equipment Name	UTID #
Balance	Sartorius Analytic A200S	407663
Oven (30°C)	Lab-Line Instruments Environ-Shaker 3597 (Stable Platform)	407675
Oven (50°C)	Precision Mechanical Convection Incubator	417018

VITA

David Greene was born in Johnson City, Tennessee in February 1967. After graduating high school, he pursued an Industrial Engineering degree at the University of Tennessee, Knoxville which he received in 1991. Upon graduation, he entered the Master of Business Administration program at the University of Tennessee, Knoxville. He graduated with an MBA in 1993.

He worked for the Tennessee Valley Authority in a variety of positions touching on engineering, environmental benefit evaluation, project finance, and internal finance. He began as a Power Supply Analyst eventually rising to a position as the Manager of Bondholder Relations in the organization's Finance department.

In 2004, he enrolled in the graduate program in the Department of Civil and Environmental Engineering at the University of Tennessee, Knoxville and accepted a graduate research assistantship from the department. He received his M.S. degree in Environmental Engineering with a concentration in Air Quality from the University of Tennessee, Knoxville in May 2005.